Proctonix AI – Smart Exam Proctoring Using AI and Webcam

CAPSTONE PROJECT REPORT

Submitted by

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DECLARATION

We affirm that the project work titled **"Proctonix AI – Smart Exam Proctoring Using AI and Webcam"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

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PROJECT SUMMARY

Project Title	Proctonix AI – Smart Exam Proctoring Using AI and Webcam	
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Guide Name/Designation	Ms. VISHNU PRIYA M.R	
Program Concentration Area	Artificial Intelligence and Computer Vision (Intelligent Systems)	
Technical Requirements	Artificial Intelligence, Computer Vision, Digital Image Processing, Python Programming, OpenCV, MediaPipe FaceMesh, Flask Framework, HTML, CSS, JavaScript, WebRTC, Screen Capture API, Web Speech API, Real-time Monitoring Systems, Behavioral Analysis using Heuristic Logic, Webcam Stream Processing, Audio Monitoring, Browser-based Application Development	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system is developed using open-source technologies such as OpenCV, MediaPipe FaceMesh, Flask, and JavaScript browser APIs, eliminating licensing costs. It operates efficiently on standard laptops with webcams and does not require high-end GPU hardware, ensuring affordability for educational institutions and online examination platforms.	✓
Environmental	The system is fully software-based and does not require specialized external hardware, thereby reducing electronic waste. It minimizes power consumption by using lightweight pre-trained models and optimized real-time processing through browser-based monitoring.	✓
Social	The framework promotes fairness and transparency in online examinations by detecting suspicious activities such as impersonation and distraction. It enhances accessibility by providing a secure and non-intrusive monitoring environment using standard devices for remote assessment.	✓

Ethical	The system adheres to ethical AI principles by ensuring student data privacy and avoiding facial recognition storage. It focuses only on facial landmark detection and behavioral monitoring such as gaze tracking and face presence. No personally identifiable information (PII) is stored.	✓
Health and Safety	The system contributes to maintaining a fair examination environment by detecting suspicious activities such as multiple face presence and prolonged distraction. It reduces academic malpractice and ensures a secure and reliable assessment process through real-time monitoring.	✓
Manufacturability	Since the project is software-oriented, deployment only requires standard computing devices with webcams and internet connectivity. The modular design allows easy integration into web-based examination systems and learning management platforms.	✓
Sustainability	The lightweight computational model ensures lower energy consumption. The system encourages long-term adoption of secure online examination practices, supporting sustainable digital education solutions.	✓

REALISTIC CONSTRAINTS

Economic Constraints:

The proposed online proctoring system is economically feasible as it relies entirely on open-source technologies including MediaPipe and OpenCV. These frameworks eliminate licensing costs associated with proprietary AI solutions. Furthermore, the system operates efficiently on standard personal computers without requiring high-end GPU infrastructure. This ensures cost-effective deployment in educational institutions and online certification platforms. The low implementation cost increases scalability and adoption potential.

Environmental Constraints:

As a software-centric solution, the project does not demand specialized hardware components, thereby reducing material consumption and electronic waste generation. The lightweight architecture minimizes energy consumption during runtime. The system's optimized processing pipeline reduces computational redundancy, contributing to lower power usage and environmentally responsible operation.

Social Constraints:

The system contributes positively to society by promoting fairness and transparency in online examinations. By enabling real-time monitoring of candidate behavior, the framework supports institutions in maintaining academic integrity during remote assessments. The accessibility of the system encourages adoption across educational environments with minimal hardware requirements.

Ethical Constraints:

The project aligns with ethical AI standards by ensuring secure handling of examination data and limiting monitoring strictly to behavior-related indicators such as gaze direction, face presence, and audio anomalies. The system avoids storing facial identity datasets and does not perform biometric authentication beyond session monitoring. Transparency mechanisms inform users about data collection processes, ensuring responsible deployment.

Applicable standards:

- IEEE 7000 – Ethical Considerations in System Design
- IEEE 2857 – Privacy Engineering in Biometric Systems

Health and Safety Constraints:

Proctonix AI enhances examination reliability while maintaining student comfort by using non-intrusive monitoring techniques. Real-time alerts help students remain attentive without interrupting exam flow. The system avoids continuous manual supervision stress by automating suspicious activity detection, improving the psychological comfort of test-takers during online assessments.

Manufacturability Constraints

Although primarily software-based, the modular system architecture allows integration into institutional examination portals and LMS platforms. The system can be deployed across multiple environments using standard browsers without additional installation requirements. This flexibility ensures ease of implementation across educational institutions with varying infrastructure capabilities.

Sustainability Constraints

The proposed system supports sustainable digital transformation in education by enabling secure remote assessments without physical infrastructure dependency. The efficient use of pre-trained computer vision models ensures reduced energy consumption during operation. By promoting long-term adoption of online examination platforms, the system contributes to environmentally responsible academic practices.

ABSTRACT

Online examinations have become increasingly common with the rapid growth of digital learning platforms; however, maintaining academic integrity in remote assessments remains a major challenge. Traditional online proctoring methods often rely heavily on manual supervision or expensive proprietary systems, which may be inefficient, intrusive, or difficult to scale. To address these limitations, this project proposes an AI-based smart online proctoring system for real-time monitoring of candidates during examinations using webcam, screen activity, and audio analysis.

The proposed system leverages computer vision techniques using OpenCV and MediaPipe FaceMesh to detect suspicious behaviors such as absence of face, presence of multiple faces, and prolonged off-screen gaze. These detections are supported by custom heuristic logic that evaluates candidate behavior in real time and generates alerts when abnormal patterns are observed. The system improves monitoring efficiency by providing automated behavioral analysis without requiring complex model training or high-end hardware.

The entire solution is implemented as a browser-based web application using Flask along with HTML, CSS, and JavaScript technologies. Students grant access to webcam, microphone, and screen activity through secure browser APIs, enabling continuous monitoring during examinations. The system automatically captures violation screenshots, generates real-time alerts, and logs suspicious events for administrative review through a dedicated proctor dashboard.

By combining computer vision, real-time monitoring, and automated reporting into a single lightweight platform, the proposed system provides an efficient and scalable solution for maintaining fairness in online examinations. It reduces reliance on manual supervision, enhances detection accuracy, and supports institutions in conducting secure remote assessments. Overall, this approach demonstrates the effectiveness of AI-driven proctoring systems in strengthening the credibility and reliability of digital education environments.

Keywords - Online Examination Monitoring, Artificial Intelligence, Computer Vision, MediaPipe FaceMesh, OpenCV, Flask Web Application, Facial Landmark Detection, Suspicious Activity Detection, Gaze Tracking, Screenshot Logging, Violation Detection, Administrative Dashboard.

TABLE OF CONTENTS

TITLE		PAGE NO.
ABSTRACT		VII
LIST OF TABLES		XI
LIST OF FIGURES		XII
LIST OF ACADEMIC REFERENCE COURSES		XIII
CHAPTER I	INTRODUCTION	1
1.1	Background and Motivation	
1.2	Problem Statement	
1.3	Objectives of the Project	
1.4	Scope of the Project	
1.5	Methodology Overview	
1.6	Organization of the Report	
CHAPTER II	LITERATURE REVIEW	5
2.1	Overview of Related Work	
2.2	Review of Similar Projects or Research Papers	
2.3	Summary and Gap Identification	
CHAPTER III	SYSTEM ANALYSIS	8
3.1	Requirements Gathering	
3.2	Functional Requirements	
3.3	Non-Functional Requirements	
3.4	Feasibility Study 3.4.1 Technical Feasibility 3.4.2 Operational Feasibility *3.4.3* Economic Feasibility	
3.5	Risk Analysis	
CHAPTER IV	SYSTEM DESIGN	12
4.1	Overall System Architecture	
4.2	Module Design	
4.2.1	Module 1	
4.2.2	Module 2	

4.3	**Database Design**	
4.3.1	**ER Diagram**	
4.3.2	**Database Schema**	
4.4	**User Interface Design**	
4.4.1	**User Flow Diagrams**	
CHAPTER V	**IMPLEMENTATION**	**18**
5.1	**Technology Stack**	
5.1.1	**Programming Languages and Tools**	
5.2	**Implementation of Modules**	
5.2.1	**Module 1**	
5.2.2	**Module 2(Repeat as many time as required)**	
5.3	**Integration of Modules**	
CHAPTER VI	**TESTING**	**22**
6.1	**Testing Methodology**	
6.1.1	**Unit Testing**	
6.1.2	**Integration Testing**	
6.1.3	**System Testing**	
6.1.4	**User Acceptance Testing (UAT)**	
6.2	**Test Cases and Results**	
6.3	**Bug Tracking and Resolution**	

CHAPTER VII	**RESULTS AND DISCUSSION**	**28**
7.1	**System Output Screenshots**	
7.2	**Evaluation Metrics**	
7.3	**Comparison with Existing Systems**	
7.4	**Challenges Faced**	
7.5	**Solutions and Improvements**	
CHAPTER VIII	**CONCLUSION & FUTURE SCOPE**	**38**
REFERENCES : List of books, articles, research papers		**39**
PUBLICATION		
CERTIFICATIONS		
PLAGIARISM REPORT		

LIST OF TABLES

TABLES	DETAILS	PAGE NO.
Table 1	Functional Test Cases	24
Table 2	Performance Evaluation Results	25
Table 3	Environment-wise Performance Evaluation	26
Table 4	Latency Measurement	26
Table 5	Bug Tracking Log	27

LIST OF FIGURES

FIGURES	DETAILS	PAGE NO.
Figure 1	Architecture of the system	13
Figure 2	Model Inference Module	14
Figure 3	Database Schema	15
Figure 4	User Flow Diagrams 1	16
Figure 5	User Flow Diagrams 2	17
Figure 6	1ntegration of Modules	21
Figure 7	Login Page Interaction Design	28
Figure 8	available job opportunities to apply	29
Figure 9	application submission for student	29
Figure 10	My job applications	30
Figure 11	Admin Dashboard	30
Figure 12	create new job posting	31
Figure 13	exam preparation	31
Figure 14	Managing applications	32
Figure 15	Email sending with exam link and password	32
Figure 16	Confusion Matrix At Different Monitoring Conditions	33
Figure 17	Performance Metrics Across Different Monitoring Conditions	35

LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	211CSE1402	Python Programming
2	212CSE2304	Machine Learning
3	212CSE2303	Software Engineering
4	213CSE3301	Deep Learning
5	213CSE1301	AI And ML

CHAPTER – I

INTRODUCTION

1.1 Background and Motivation

Online examination systems have become an essential component of modern digital education platforms, especially with the rapid growth of remote learning environments. While these systems provide flexibility and accessibility to students, maintaining academic integrity during remote assessments remains a major challenge. Traditional proctoring methods rely heavily on manual supervision or physical presence, which becomes impractical in large-scale online examinations. With advancements in artificial intelligence and computer vision, real-time automated proctoring has emerged as an efficient solution for monitoring candidate behavior during examinations.

Recent developments in computer vision technologies such as OpenCV and MediaPipe FaceMesh have significantly improved the ability to detect facial landmarks and analyze user attention from webcam video streams. These frameworks enable accurate face detection, gaze tracking, and presence monitoring in real time using lightweight models. However, many existing online proctoring solutions still face challenges related to scalability, cost, privacy concerns, and computational complexity. Most implementations depend on expensive proprietary platforms or require high-end hardware, which limits their accessibility in educational institutions.

The motivation behind this project is to design a real-time intelligent online proctoring system that detects suspicious behaviors such as absence of face, multiple face presence, and prolonged off-screen gaze using computer vision techniques. By combining MediaPipe FaceMesh with OpenCV-based video processing and browser-based monitoring technologies, the proposed framework aims to bridge the gap between traditional manual invigilation and automated remote examination monitoring systems.

1.2 Problem Statement

With the rapid transition toward online education, ensuring fairness and credibility in remote examinations has become increasingly difficult. Traditional online assessment platforms lack effective monitoring mechanisms to detect suspicious activities such as impersonation, use of unauthorized resources, and candidate distraction during examinations. Manual monitoring through video conferencing tools is inefficient and impractical for large numbers of students.

Although several AI-based proctoring systems have been developed, many of them require expensive infrastructure, complex installation procedures, or high computational resources. Additionally, some systems raise privacy concerns by storing biometric identity information or continuous surveillance recordings. Lightweight monitoring solutions often fail to provide reliable behavioral detection, reducing their effectiveness in maintaining examination integrity.

Therefore, there is a need for an efficient and scalable AI-based online proctoring system that can monitor candidates in real time using standard webcams and browser technologies. The system should detect suspicious activities accurately while maintaining user privacy and operating efficiently on standard computing devices. Such a solution can improve examination transparency, reduce malpractice, and support secure digital learning environments.

1.3 Objectives of the Project

The primary objective of this project is to develop an intelligent and efficient system for monitoring online examinations using computer vision and artificial intelligence techniques. The system aims to detect suspicious candidate behavior in real time through webcam, audio, and screen activity monitoring.

Another important objective is to implement facial landmark detection using MediaPipe FaceMesh along with OpenCV to identify behaviors such as face absence, multiple face presence, and prolonged off-screen gaze. The system also focuses on improving monitoring accuracy while ensuring lightweight performance suitable for standard computing devices.

The project aims to design and develop a browser-based web application using Flask, HTML, CSS, and JavaScript that allows students to participate in monitored examinations without installing additional software. The system also provides real-time alerts and automatically captures screenshots when suspicious activities are detected.

Additionally, the project focuses on generating violation logs and reports that assist administrators in reviewing examination sessions efficiently. The overall objective is to create a scalable, cost-effective, and reliable solution that supports institutions in maintaining academic integrity during online assessments.

1.4 Scope of the Project

The scope of this project focuses on the development of an AI-based online examination monitoring system using webcam, screen capture, and audio analysis technologies. The system is designed to assist educational institutions in detecting suspicious candidate behavior during remote examinations and maintaining examination transparency.

The project includes the implementation of computer vision techniques using OpenCV and MediaPipe FaceMesh for facial landmark detection and gaze tracking. It also incorporates browser-based APIs such as WebRTC and Screen Capture API for monitoring webcam activity and screen usage during examinations.

The system is developed as a web-based application using Flask, allowing students to participate in monitored examinations through standard browsers without installing additional software. It supports automated alert generation and screenshot logging for suspicious events detected during examination sessions.

The scope also covers real-time behavioral monitoring, violation logging, and administrative dashboard support for reviewing candidate activity. The system is designed to operate efficiently on standard computing devices, ensuring accessibility and scalability across educational institutions.

However, the system is intended to act as a supportive monitoring tool rather than a replacement for human invigilators. Future enhancements may include integration with learning management systems, advanced behavioral analysis using deep learning models, and deployment on cloud-based examination platforms.

Overall, the project aims to provide a practical, efficient, and scalable solution that bridges the gap between traditional invigilation methods and intelligent automated online examination monitoring systems.

1.5 Methodology Overview:

The proposed system follows a structured methodology to develop an efficient and reliable AI-based solution for monitoring online examinations using webcam, audio, and screen activity analysis. The methodology begins with user authentication, where candidates log into the examination portal through a browser-based interface and grant permission for accessing webcam, microphone, and screen capture features. These inputs serve as the primary data sources for monitoring candidate behavior during the examination session. The system ensures secure initialization of monitoring components before the examination begins to maintain a controlled assessment environment.

The next step involves real-time video acquisition and processing using WebRTC and OpenCV. The captured webcam frames are analyzed continuously to detect facial presence and head orientation. MediaPipe FaceMesh is used to identify detailed facial landmarks, enabling accurate tracking of eye movement and gaze direction. Additional preprocessing steps such as frame resizing and landmark normalization are performed to improve detection efficiency and ensure stable performance across different lighting conditions and camera qualities.

Following preprocessing, the system utilizes MediaPipe FaceMesh-based landmark detection along with custom heuristic logic to analyze candidate behavior. These heuristics evaluate conditions such as absence of face, presence of multiple faces in the frame, and prolonged off-screen gaze. When abnormal behavior is detected, the system generates alerts indicating possible suspicious activity. This rule-based evaluation approach ensures efficient monitoring without requiring complex model training or large datasets.

Once suspicious activity is identified, the system automatically captures timestamped screenshots and logs violation events for administrative review. The monitoring process also includes screen activity tracking using the Screen Capture API to ensure that candidates remain within the examination environment. Audio monitoring is performed to detect unusual background noise or conversations that may indicate unauthorized assistance during the examination session.

The system is integrated into a web-based application developed using Flask, HTML, CSS, and JavaScript. The application provides a user-friendly interface for students to participate in examinations and enables administrators to monitor candidate activity through a dedicated dashboard. Real-time alerts and violation logs are displayed to help proctors take appropriate action when required.

Additionally, the methodology includes automated report generation that records suspicious activity detected during the examination session. These reports contain captured screenshots, timestamps, and behavioral analysis summaries, which assist administrators in reviewing examination integrity effectively. The modular architecture ensures seamless communication between frontend monitoring components and backend processing modules.

The methodology ensures efficient handling of multimedia data streams, accurate behavioral analysis, and smooth integration of computer vision techniques with a browser-based monitoring interface. The system is designed to operate on standard computing devices without requiring specialized hardware, ensuring accessibility and scalability for educational institutions.

Finally, the entire workflow is tested under different environmental conditions to ensure robustness, reliability, and consistent monitoring performance. The methodology bridges the gap between traditional manual invigilation methods and intelligent automated proctoring

systems by combining real-time monitoring, behavioral analysis, and web-based deployment into a single integrated solution.

1.6 Organization of the Report:

This report is organized into eight major chapters, each addressing a specific phase of the project development lifecycle, from conceptual foundation to implementation, evaluation, and conclusion.

Chapter I – Introduction provides the background and motivation behind AI-based online proctoring systems, outlines the problem statement, defines the objectives and scope of the project, and gives an overview of the methodology adopted. This chapter highlights the importance of maintaining academic integrity in remote examinations using artificial intelligence.

Chapter II – Literature Review examines existing research related to online proctoring systems, computer vision-based monitoring techniques, and AI applications in remote education. It reviews traditional invigilation methods, automated behavior detection techniques, and identifies research gaps in lightweight real-time monitoring solutions.

Chapter III – System Analysis presents the functional and non-functional requirements of the system, including webcam monitoring, gaze detection, screen tracking, and audio analysis. This chapter also includes feasibility analysis covering technical, operational, and economic aspects, along with risk analysis related to privacy concerns and detection reliability.

Chapter IV – System Design explains the overall architecture of the proposed system, including webcam input processing, facial landmark detection using MediaPipe FaceMesh, behavior analysis, alert generation, and violation logging modules. It also presents workflow diagrams and interface design of the monitoring application.

Chapter V – Implementation describes the technologies used in system development such as Flask, OpenCV, MediaPipe FaceMesh, HTML, CSS, and JavaScript. It explains the implementation of modules including webcam monitoring, gaze tracking, audio monitoring, screenshot capture, and dashboard integration.

Chapter VI – Testing presents the testing strategies used to validate system performance through unit testing, integration testing, and system testing. Test cases verify monitoring accuracy, alert generation, and violation logging functionality.

Chapter VII – Results and Discussion analyzes system performance under different conditions and evaluates the effectiveness of facial landmark detection, gaze tracking accuracy, and real-time alert mechanisms.

Chapter VIII – Conclusion and Future Scope summarizes the achievements of the project and discusses future enhancements such as LMS integration, advanced behavior recognition, and cloud-based deployment for improved scalability.

Finally, the report concludes with the References section listing all academic sources consulted, followed by documentation of Publications, Certifications, and the Plagiarism Report as required by institutional guidelines.

CHAPTER-II
LITERATURE REVIEW

2.1 Overview of Related Work

Online examination monitoring has gained significant research attention due to the rapid growth of remote learning environments and digital education platforms. Traditional invigilation approaches rely on physical supervision in examination halls, which ensures controlled environments but becomes impractical in large-scale online assessments. With the advancement of artificial intelligence and computer vision technologies, automated online proctoring systems have emerged as an effective solution for maintaining academic integrity during remote examinations.

Initial research in this domain focused on webcam-based monitoring systems that detect face presence and candidate activity using classical image processing techniques. These systems demonstrated the feasibility of automated supervision but faced limitations in accuracy and scalability. Over time, more advanced frameworks such as OpenCV and MediaPipe FaceMesh have been introduced, significantly improving real-time facial landmark detection and gaze tracking performance.

Recent studies have also emphasized the integration of behavioral analysis techniques in online proctoring systems. Methods such as head pose estimation, gaze tracking, and multiple-face detection are widely used to identify suspicious examination behavior. These techniques help monitor candidate attention levels and reduce the possibility of impersonation or external assistance during examinations.

Furthermore, the development of browser-based monitoring solutions has enabled real-time deployment of intelligent proctoring systems without requiring specialized hardware installations. Frameworks such as Flask allow seamless integration of artificial intelligence models into web-based platforms where administrators can monitor candidate activity efficiently. These advancements highlight the shift toward scalable, accessible, and privacy-aware online examination monitoring solutions.

2.2 Review of Similar Projects or Research Papers:

A systematic review on artificial intelligence in online proctoring systems classified monitoring approaches into behavioral analysis, computer vision techniques, and audio-based detection methods. The study highlighted the effectiveness of multimodal monitoring in improving examination integrity while identifying privacy-related concerns in automated surveillance systems. It also emphasized the importance of balancing security requirements with transparency and ethical data handling in large-scale online assessment environments. [1]

A real-time cheating detection system based on YOLO demonstrated the capability to detect suspicious objects such as mobile phones and additional persons during online examinations. The approach improved detection accuracy through deep learning-based object recognition techniques but required higher computational resources for deployment. This highlighted the trade-off between detection performance and system efficiency in real-time monitoring applications. [2]

A multi-modal online proctoring framework integrating face detection, gaze tracking, and audio monitoring achieved reliable performance in identifying suspicious candidate behavior. The system demonstrated improved monitoring accuracy compared to single-modality approaches by combining multiple behavioral indicators. Such integration significantly enhanced the robustness of automated examination supervision systems. [3]

Research on student acceptance of AI-based online proctoring systems emphasized the importance of transparency and user trust in monitoring platforms. The study identified privacy

awareness and communication as key factors influencing adoption of automated proctoring technologies. It further suggested that clearly informing users about monitoring procedures can improve acceptance and reduce resistance toward AI-based supervision systems. [4]

A behavioral monitoring framework using facial landmark detection techniques demonstrated improved accuracy in identifying distraction patterns during remote examinations. The system effectively tracked gaze direction and head movement using lightweight computer vision models suitable for real-time applications. This approach enabled scalable monitoring without requiring expensive hardware infrastructure. [5]

Studies on keystroke dynamics for continuous authentication showed that typing behavior analysis can support identity verification during online assessments. The technique provided an additional security layer by analyzing typing rhythm patterns throughout the examination session. However, the approach was limited to text-based examinations and required additional training data for reliable performance. [6]

A red-teaming analysis of online proctoring platforms identified vulnerabilities such as virtual camera spoofing and screen mirroring attacks. The study recommended integrating multiple monitoring techniques to improve system reliability and reduce security risks. It also highlighted the importance of designing resilient proctoring architectures capable of handling emerging cheating strategies. [7]

Research on deep learning-based gaze tracking systems demonstrated the effectiveness of facial landmark detection models in identifying candidate attention levels. These systems improved monitoring performance while maintaining real-time responsiveness using optimized neural network architectures. The study confirmed that gaze tracking plays a critical role in detecting behavioral anomalies during examinations. [8]

Browser-based monitoring frameworks using WebRTC technology enabled efficient streaming of webcam and screen data without additional software installation. These systems improved accessibility and scalability of online proctoring solutions across different devices and operating systems. The approach also simplified deployment in educational institutions with limited technical infrastructure. [9]

Hybrid monitoring architectures combining computer vision and heuristic rule-based analysis achieved reliable detection of suspicious examination activities. The study demonstrated that lightweight models can provide effective monitoring without requiring complex training datasets. Such systems offered a practical balance between computational efficiency and behavioral detection accuracy. [10]

2.3 Summary and Gap Identification:

The reviewed literature demonstrates substantial progress in AI-based online proctoring systems using computer vision, behavioral monitoring, and multimodal detection techniques. Many systems successfully achieve real-time monitoring accuracy using facial landmark detection, gaze tracking, and object detection frameworks. Furthermore, integration of browser-based monitoring technologies has improved accessibility and scalability of remote examination platforms.

However, several research gaps remain evident. First, many existing systems rely on computationally intensive deep learning models that require high-performance hardware, limiting deployment in resource-constrained environments. Second, some monitoring solutions focus primarily on object detection rather than continuous behavioral analysis using facial landmarks and gaze tracking techniques.

Third, privacy concerns remain a major challenge in automated proctoring systems, as several platforms store biometric identity data or continuous video recordings. Additionally, limited attention has been given to lightweight browser-based monitoring frameworks that operate efficiently without specialized installations. Finally, integration of real-time alert mechanisms and automated violation logging features is not sufficiently explored in many earlier implementations.

The proposed system addresses these gaps by integrating MediaPipe FaceMesh-based facial landmark detection with OpenCV video processing and browser-based monitoring technologies for real-time behavior analysis. Unlike previous approaches that rely heavily on complex deep learning pipelines, this framework emphasizes lightweight deployment, privacy-aware monitoring, and automated alert generation. By combining accessibility, computational efficiency, and real-time detection capabilities, the proposed system provides a scalable and practical solution for intelligent online examination monitoring.

CHAPTER III

SYSTEM ANALYSIS

3.1 Requirements Gathering

The requirements for the proposed Proctonix AI online proctoring system were identified through systematic evaluation of institutional needs, examination monitoring challenges, and technical constraints associated with remote assessment environments. The primary stakeholders considered during the requirement analysis include educational institutions, examination administrators, instructors, and students participating in online examinations. The system is designed to assist administrators in monitoring candidate behavior effectively by detecting suspicious activities through webcam, screen activity, and audio monitoring technologies, thereby improving examination transparency and reliability.

The requirements gathering process involved analyzing existing AI-based online proctoring systems, reviewing literature on computer vision-based behavioral monitoring techniques, and identifying limitations in current remote invigilation solutions. Special emphasis was placed on achieving reliable real-time monitoring, efficient processing speed, and secure handling of examination data. Lightweight computer vision frameworks such as OpenCV and MediaPipe FaceMesh were selected to ensure accurate facial landmark detection and gaze tracking without requiring high computational resources.

Additionally, the system is designed to function as a browser-based application using Flask along with HTML, CSS, and JavaScript technologies, enabling students to participate in monitored examinations without installing additional software. The system also incorporates heuristic-based behavioral analysis techniques to detect suspicious activities such as absence of face, presence of multiple faces, and prolonged off-screen gaze during examination sessions.

Key considerations during requirements analysis include monitoring accuracy, processing latency, usability of the student interface, effectiveness of the administrative dashboard, data privacy protection, and system scalability for monitoring multiple candidates simultaneously. The system must also support automated alert generation, screenshot capture, and violation logging to assist administrators in reviewing examination integrity efficiently. Furthermore, it is designed to operate effectively on standard computing devices with integrated webcams and microphones, ensuring accessibility and cost-effective deployment across educational institutions.

These requirements collectively guided the design and development of the proposed system, ensuring a balance between monitoring performance, usability, privacy protection, and practical deployment in real-world online examination environments.

3.2 Functional Requirements

The functional requirements define the specific operations that the system must perform to achieve its intended objectives. The system must be capable of capturing real-time webcam video input through browser-based access and processing individual frames continuously without noticeable delay. It must accurately detect facial landmarks using MediaPipe FaceMesh and analyze candidate head orientation and gaze direction during examination sessions.

The system must identify suspicious behaviors such as absence of face from the frame, detection of multiple faces in the camera view, and prolonged off-screen gaze that may indicate distraction or external assistance. When such activities are detected, the system must generate real-time alerts and record violation events for administrative review. Additionally, the system must capture timestamped screenshots automatically whenever suspicious activity occurs.

The framework must also support screen activity monitoring using browser-based screen capture APIs to ensure that candidates remain within the examination environment. Audio monitoring capability should be included to detect unusual background noise or conversations that may indicate unauthorized communication during the examination session.

Furthermore, the system must provide an administrative dashboard that allows proctors to monitor multiple candidates simultaneously and review violation logs efficiently. The framework should maintain stable monitoring performance under varying lighting conditions and camera qualities while ensuring continuous operation throughout the examination session.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes that the system must satisfy to ensure reliable performance. One of the most critical non-functional requirements is real-time processing capability. The system must analyze webcam frames continuously with minimal latency to maintain effective monitoring throughout the examination session.

Accuracy is another essential requirement. The system should reliably detect facial landmarks and suspicious behavioral patterns such as multiple-face presence and off-screen gaze under varying environmental conditions. Stability under moderate lighting variations and different camera resolutions is also important for ensuring consistent monitoring performance.

Scalability is required to allow future expansion toward monitoring multiple candidates simultaneously through centralized administrative dashboards. The system must follow a modular architecture that enables independent modification of components such as video processing, behavioral analysis, and alert generation modules.

Security and privacy must also be maintained. The system should avoid storing unnecessary personal data and ensure secure handling of monitoring information during examination sessions. Finally, usability and interface clarity are essential to ensure that both students and administrators can interact with the system easily without requiring technical expertise.

3.4 Feasibility Study

A feasibility study was conducted to evaluate whether the proposed Proctonix AI online proctoring system can be successfully implemented within technical, operational, and economic constraints. The analysis confirms that the system is feasible due to the availability of open-source computer vision technologies, browser-based monitoring frameworks, and standard computing infrastructure.

From a technical perspective, the system is feasible as it utilizes widely adopted frameworks such as OpenCV, MediaPipe FaceMesh, and Flask. These tools support efficient real-time video processing, facial landmark detection, and web-based deployment. The use of lightweight pre-trained models further simplifies implementation and ensures reliable monitoring performance without requiring specialized hardware resources.

From an operational perspective, the system is user-friendly and can be easily adopted by educational institutions conducting online examinations. It provides a simple interface for students to participate in monitored sessions and allows administrators to review suspicious activity logs efficiently through a centralized dashboard.

From an economic perspective, the system is cost-effective as it relies entirely on open-source software technologies and standard computing devices. No specialized monitoring equipment or

additional infrastructure is required for deployment, making the system accessible for institutions with limited technical resources.

Overall, the feasibility study indicates that the proposed system is practical, scalable, and suitable for real-world deployment in online examination environments.

3.4.1 Technical Feasibility

The technical feasibility of the system is strong due to the availability of mature open-source libraries such as OpenCV and MediaPipe FaceMesh. MediaPipe provides optimized facial landmark detection capable of real-time performance on standard hardware, while OpenCV supports efficient video capture and image processing operations.

The computational requirements are moderate, and the system can operate on personal computers equipped with standard processors and integrated webcams. Optional GPU acceleration can further enhance processing performance but is not mandatory for system operation. The heuristic-based behavioral detection logic is computationally lightweight, making implementation technically practical.

Furthermore, the modular system architecture allows independent development and testing of components such as webcam monitoring, screen capture integration, and alert generation modules, reducing implementation complexity. Therefore, from a technical standpoint, the project is feasible and implementable using currently available technologies.

3.4.2 Operational Feasibility

Operational feasibility evaluates whether the system can function effectively in real-world online examination environments and whether users can easily adopt it. The proposed system is designed as a browser-based monitoring application that requires only a standard computer or laptop with webcam and internet connectivity, making it suitable for use across educational institutions conducting remote examinations.

The user interface is designed to be simple, intuitive, and accessible, allowing students to participate in monitored examinations without installing additional software. Administrators can monitor candidate activity through a centralized dashboard that displays alerts and violation logs in real time.

Additionally, the system integrates multiple monitoring functionalities such as webcam tracking, gaze detection, screen monitoring, and screenshot logging within a single platform. This integration reduces manual supervision effort and improves administrative efficiency during examinations.

The system can operate under varying environmental conditions such as lighting differences and camera quality variations, ensuring reliable monitoring performance across different examination settings. Therefore, the proposed system demonstrates strong operational feasibility by providing an efficient and practical solution for real-time online examination monitoring.

3.4.3 Economic Feasibility

The proposed system is economically viable due to its reliance on open-source software frameworks and minimal hardware requirements. Technologies such as OpenCV, MediaPipe FaceMesh, and Flask are freely available, eliminating the need for costly software licenses.

The system requires only a standard computing device with webcam support for deployment, which is commonly available in educational institutions. No specialized surveillance equipment or additional infrastructure is required, reducing initial investment and operational expenses.

Development costs are primarily limited to software implementation, testing, and deployment activities. Since the system is fully software-based, there are no manufacturing or hardware maintenance costs involved. Additionally, the modular architecture allows easy upgrades and scalability without significant financial overhead.

As a result, the proposed system is cost-effective and suitable for deployment across schools, colleges, universities, and certification platforms conducting remote examinations.

3.5 Risk Analysis

Although the proposed system is feasible, certain risks must be considered during implementation. One potential risk is reduced monitoring accuracy due to poor lighting conditions, low camera resolution, or unstable internet connectivity during examination sessions. These factors may affect facial landmark detection performance and behavioral analysis accuracy. This risk can be mitigated by providing system usage guidelines and implementing adaptive preprocessing techniques.

Another risk involves increased processing latency when monitoring multiple candidates simultaneously. This may impact real-time alert generation performance. The issue can be addressed through optimized processing pipelines and efficient resource management techniques.

Variability in candidate environments such as background noise or movement outside the camera frame may also affect monitoring reliability. This risk can be reduced by integrating multimodal monitoring techniques including audio detection and screen activity tracking.

Privacy concerns represent another important risk factor, as the system processes webcam and screen data during examinations. This can be mitigated by avoiding unnecessary storage of personal information and ensuring secure handling of monitoring data through controlled access mechanisms.

Hardware limitations may affect system performance on low-end devices, particularly during continuous monitoring sessions. This issue can be addressed by using lightweight computer vision models that operate efficiently on standard computing systems.

Overall, the identified risks are manageable and can be effectively mitigated through careful system design, optimization strategies, and adherence to best practices in AI-based online examination monitoring systems.

CHAPTER IV

SYSTEM DESIGN

4.1 Overall System Architecture:

The overall architecture of the proposed Proctonix AI online proctoring system follows a layered and modular design approach to ensure scalability, maintainability, and real-time monitoring efficiency. The architecture consists of five primary layers: Input Acquisition Layer, Video Processing Layer, Behavioral Analysis Layer, Alert & Logging Layer, and Output & Monitoring Layer.

The **Input Acquisition Layer** is responsible for capturing webcam video, microphone audio, and screen activity from the student's device through browser-based APIs such as WebRTC and Screen Capture API. The system validates permissions securely before initiating monitoring operations during the examination session.

The **Video Processing Layer** performs frame extraction and preprocessing using OpenCV. Each frame is analyzed continuously to detect facial landmarks and head orientation. MediaPipe FaceMesh is used to identify detailed facial landmark coordinates required for gaze tracking and presence detection.

The **Behavioral Analysis Layer** evaluates candidate activity using heuristic logic applied to facial landmark data. It detects suspicious behaviors such as absence of face, presence of multiple faces in the frame, and prolonged off-screen gaze. These detections help identify possible academic misconduct during examination sessions.

The **Alert & Logging Layer** generates real-time alerts whenever suspicious behavior is detected. It captures timestamped screenshots and records violation events for administrative review. The system maintains structured logs to assist proctors in analyzing candidate activity effectively.

The **Output & Monitoring Layer** presents monitoring results through an administrative dashboard developed using Flask. It displays violation alerts, candidate activity summaries, and captured screenshots. This layer ensures efficient supervision of multiple candidates simultaneously during examination sessions.

This modular architecture ensures efficient data flow, real-time monitoring performance, and easy scalability, making the system suitable for deployment in practical online examination environments.

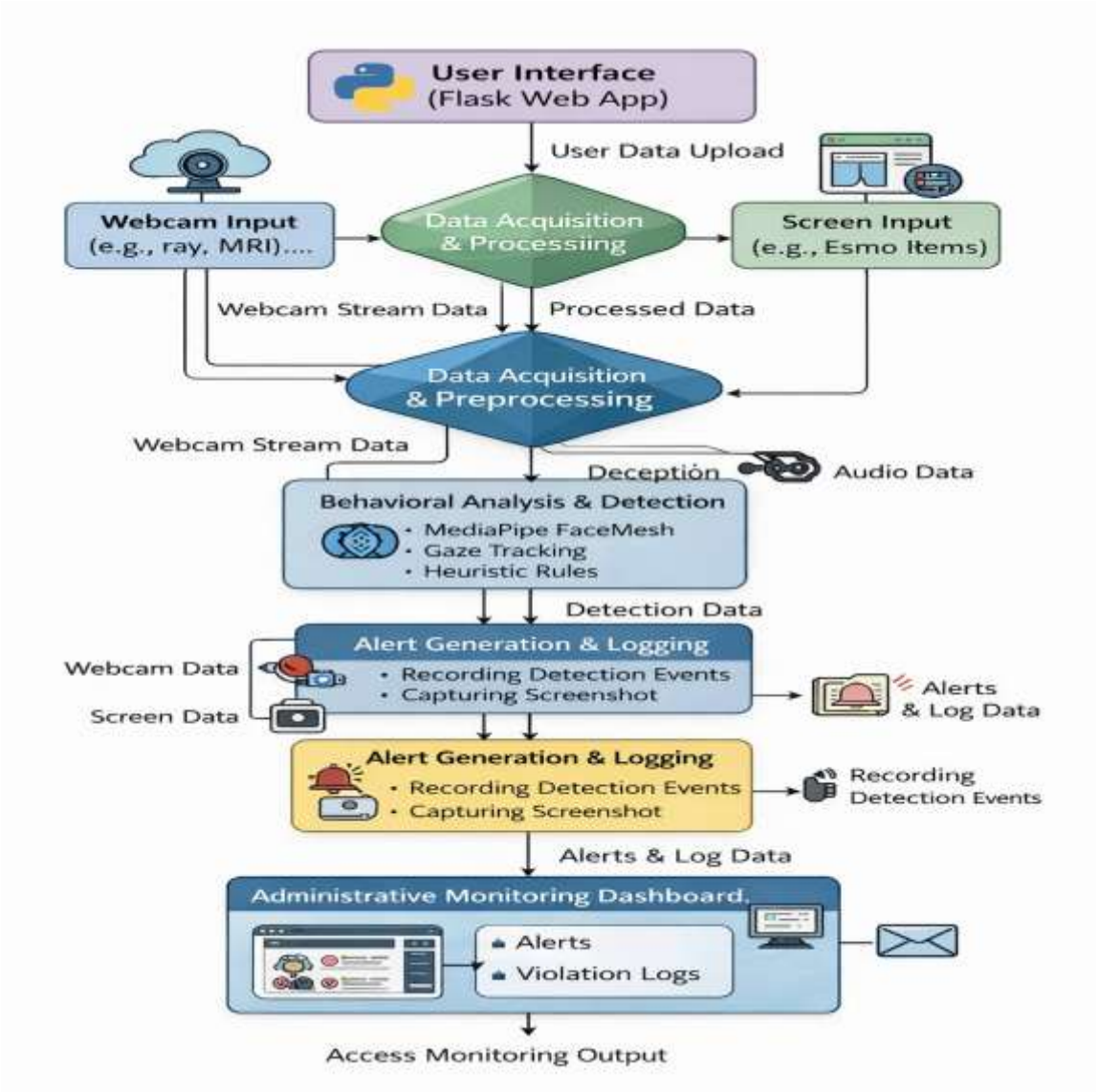

Fig.1. Architecture of the system

4.2 Module Design

The system is divided into independent modules to ensure modularity and simplified debugging. Each module performs a specific task while communicating with other modules through defined interfaces.

4.2.1 Module 1 – Model Inference Module

The Behavioral Detection Module is responsible for analyzing webcam video frames and identifying suspicious candidate activities during examinations. It utilizes MediaPipe FaceMesh and OpenCV to detect facial landmarks and evaluate head movement and gaze direction.

The module takes processed video frames as input and produces detection results such as face absence alerts, multiple face detection warnings, and prolonged off-screen gaze identification. These outputs help administrators monitor candidate behavior effectively and ensure examination integrity.

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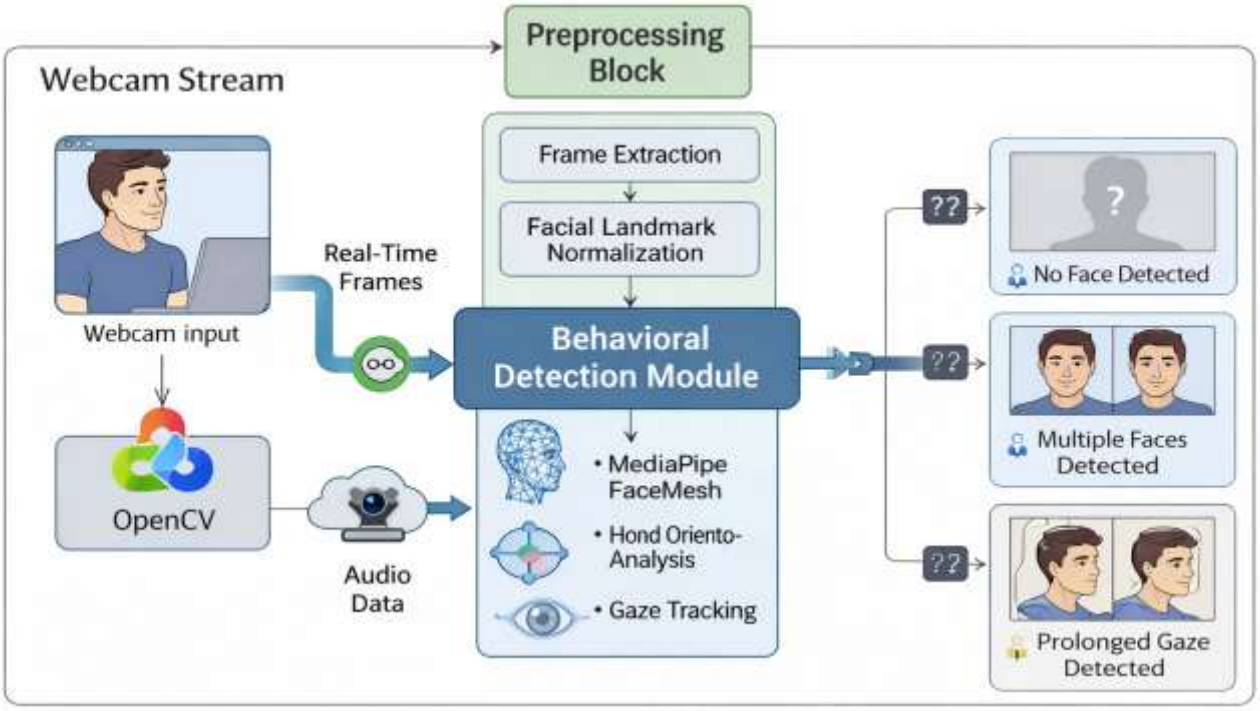

Fig.2. Model Inference Module

4.2.2 Module 2 – Image Preprocessing Module

This module processes webcam video streams captured from the browser and prepares them for behavioral analysis. It performs operations such as frame extraction, resizing, and landmark normalization using OpenCV.

The module ensures consistency in input frames and improves monitoring accuracy by handling variations caused by lighting conditions, camera resolution, and environmental factors. It standardizes the video stream format, making it compatible with MediaPipe FaceMesh detection algorithms.

4.2.3 Module 3 – Explainability (Grad-CAM) Module

This module analyzes detection results and generates alerts whenever suspicious activity is identified. It records violation events and captures screenshots automatically with timestamps for administrative review.

This module transforms the system from a simple monitoring interface into an intelligent proctoring tool by providing automated alerts that assist proctors in identifying examination irregularities efficiently.

4.2.4 Module 4 – Report Generation & Monitoring Module

This module records monitoring outputs and generates structured violation logs based on detected suspicious activities. It includes key monitoring indicators such as face presence status, gaze deviation alerts, and multiple face detection warnings.

The module stores captured screenshots and violation timestamps for later review by administrators. It also supports monitoring system performance and maintaining structured examination activity records for evaluation purposes.

4.3 Database Design :

Although the system primarily performs real-time behavioral monitoring, a lightweight data

storage mechanism is used for managing candidate activity logs, captured screenshots, and monitoring alerts. The system stores violation records, session timestamps, and administrator review data for analysis purposes.

The design avoids storing unnecessary personal biometric data permanently, ensuring privacy and security. File-based storage or minimal database integration is used to maintain simplicity, efficiency, and easy retrieval of monitoring reports when required.

4.3.1 ER Diagram:

The Entity-Relationship (ER) diagram represents the logical structure of the system designed to store candidate activity data, monitoring alerts, and violation reports for the online proctoring framework.

User – Represents a student participating in the monitored examination session.

Session – Represents an examination session associated with each user login.

Detection Log – Stores monitoring outputs generated during the examination session, including suspicious activity alerts.

Screenshot – Represents captured evidence images recorded during violation detection events.

Report – Represents generated monitoring summaries containing session details and violation records.

Relationships:

- **User participates in Session** – A user can participate in multiple examination sessions.
- **Session generates Detection Log** – Each session produces multiple monitoring alerts.
- **Detection Log generates** Screenshot – Each suspicious event may produce captured evidence images.
- **Session generates Report** – Each examination session produces a structured monitoring summary report.

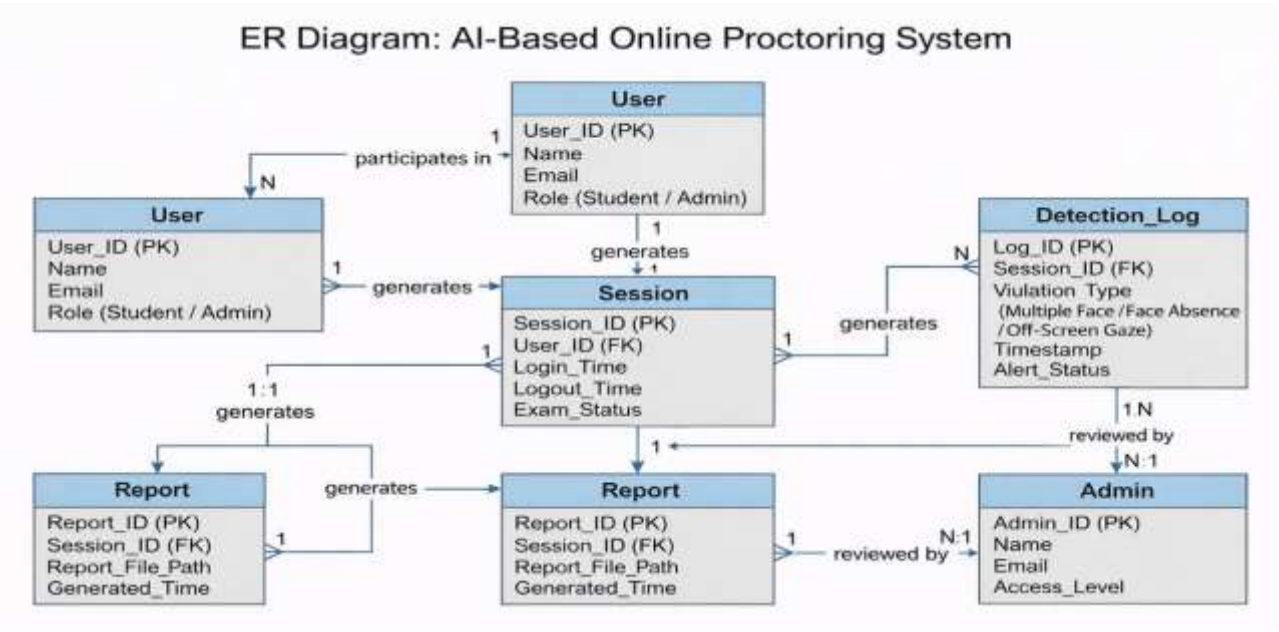

Fig.3.Database Schema

The database schema defines the physical structure of tables and their attributes required to store and manage monitoring data efficiently.

User – Stores student identification details such as username and session credentials.

Session Data – Stores examination session timestamps and monitoring duration.

Detection Logs – Stores suspicious activity alerts including face absence, multiple face detection, and gaze deviation events.

Screenshot Data – Stores captured violation image paths and related timestamps.

Report Data – Stores generated monitoring summary report file paths and associated metadata.

4.4 User Interface Design

The user interface of the proposed system is designed to ensure clarity, simplicity, and ease of interaction for both students and administrators. The primary objective of the interface is to provide secure examination monitoring while maintaining minimal distraction for users during examination sessions. The main interface consists of a login section, permission request panel for webcam and screen access, and a monitoring status display panel.

Once monitoring begins, the system processes webcam video streams and displays real-time status indicators related to face detection and gaze tracking. The administrative interface presents alerts and violation logs, highlighting suspicious behaviors detected during examination sessions. A monitoring dashboard is included to inform administrators about candidate activity status, making the output meaningful for supervision purposes.

The interface uses clean layouts and structured output sections to enhance usability. Clear labels, organized alert notifications, and easy navigation ensure that administrators can interact with the system efficiently. The system also provides options to review captured screenshots and violation logs for further analysis.

The interface ensures fast response time and smooth interaction, maintaining system reliability and user engagement.

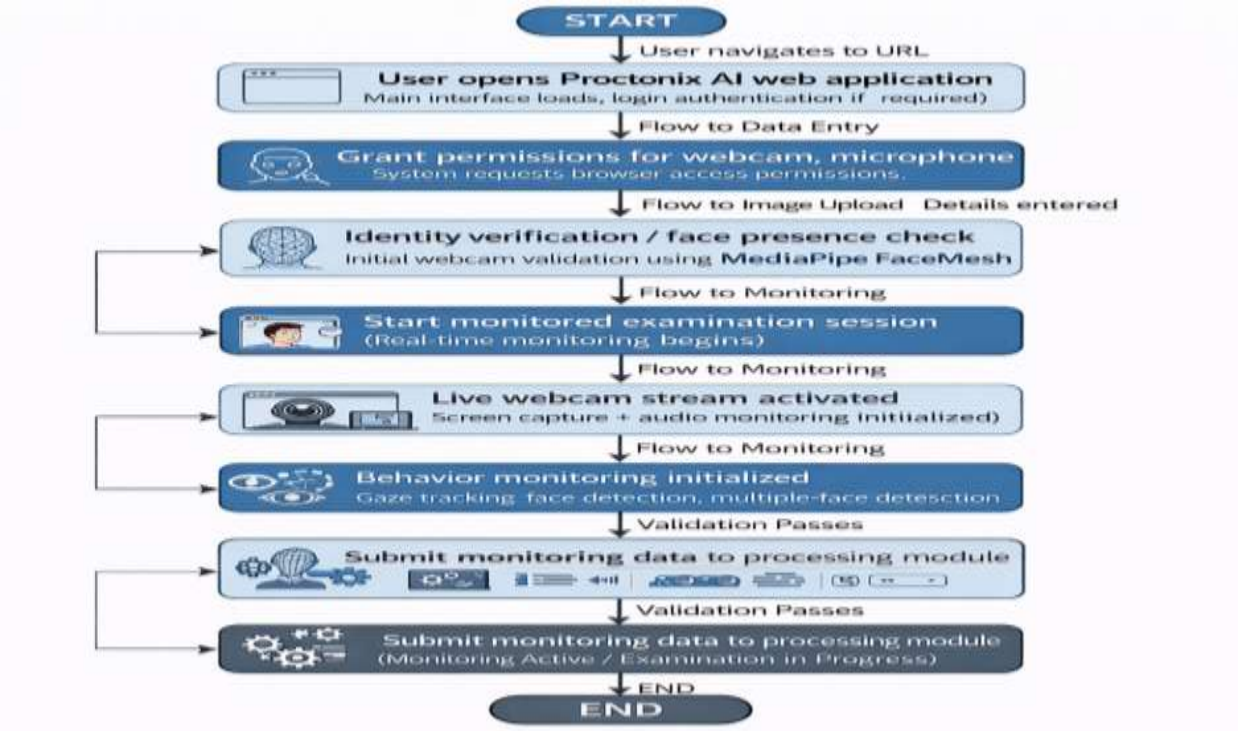

Fig.4. User Flow Diagram 1 (Input Flow)

This diagram illustrates the initial interaction flow where the student accesses the Proctonix AI web application, grants webcam, microphone, and screen permissions, completes identity verification through face presence detection, and begins the monitored examination session for real-time behavior analysis.

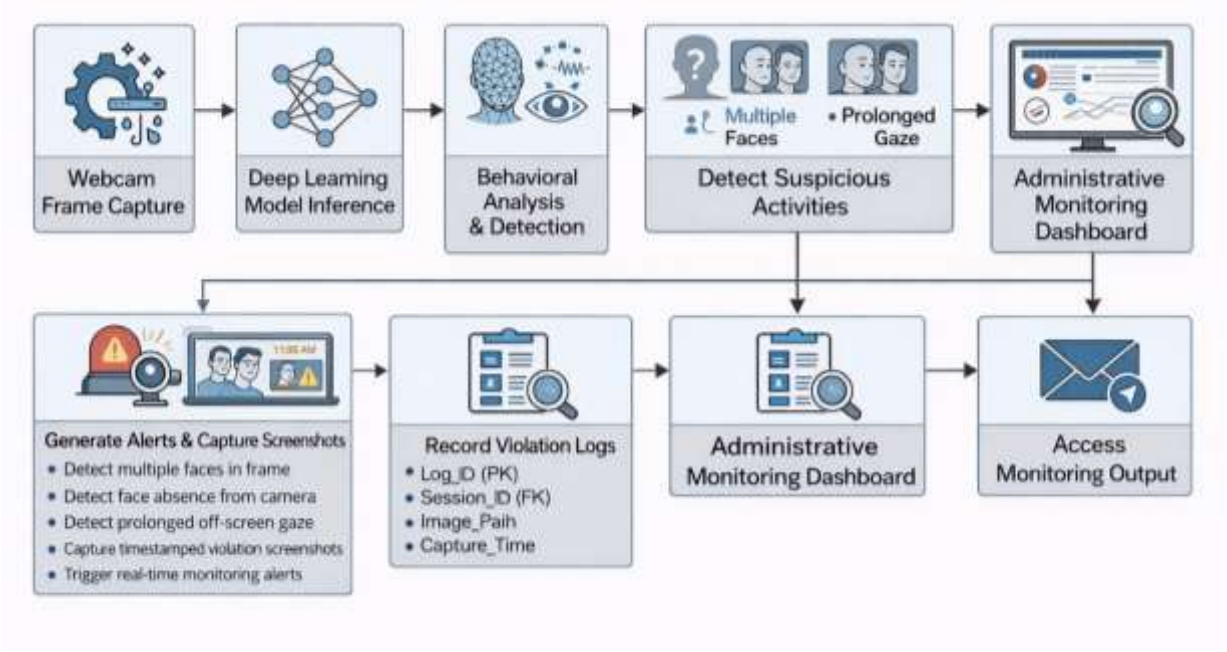

Fig.5. User Flow Diagram 2 (Processing & Output Flow)

This diagram represents the processing and output flow of the Proctonix AI system, including webcam frame capture, facial landmark detection using MediaPipe FaceMesh, behavioral analysis for identifying suspicious activities such as multiple-face presence and prolonged off-screen gaze, alert generation with screenshot capture, violation log recording, and display of monitoring results through the administrative dashboard**.**

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CHAPTER V

IMPLEMENTATION

5.1 Technology Stack:

The implementation of the proposed Proctonix AI online proctoring system is carried out using a lightweight, scalable, and efficient software stack focused on real-time monitoring and browser-based deployment. The system integrates computer vision frameworks, web technologies, and multimedia processing libraries to ensure smooth operation during online examination sessions.

The core processing engine of the system is implemented in Python, which provides strong support for computer vision, real-time video frame processing, and backend web development. Python's modular ecosystem enables seamless integration of MediaPipe, OpenCV, and Flask for building intelligent monitoring applications.

For facial landmark detection and behavioral monitoring, MediaPipe FaceMesh is used to extract facial feature coordinates from webcam video streams. These landmarks help identify suspicious behaviors such as multiple-face presence, face absence, and prolonged off-screen gaze during examinations.

For video processing operations such as frame capture, resizing, and visualization overlays, OpenCV is utilized. Additional libraries such as NumPy are used for numerical computations and heuristic-based behavioral evaluation.

The web interface is developed using Flask along with HTML, CSS, and JavaScript, enabling the system to operate as a browser-based application without requiring additional installation. Browser APIs such as WebRTC and Screen Capture API are integrated to enable webcam monitoring and screen activity tracking during examination sessions.

The system also includes automated screenshot logging and violation record storage modules that support administrative review of examination activity. This technology stack ensures platform independence, scalability, and efficient performance on standard computing devices.

5.1.1 Programming Languages and Tools

The primary programming language used in the project is Python, chosen for its strong ecosystem supporting artificial intelligence, real-time video processing, and backend web application development. Python simplifies implementation of monitoring algorithms and integration with browser-based interfaces.

The key tools and libraries used in the implementation are:

- **MediaPipe FaceMesh** – For facial landmark detection and gaze tracking.
- **OpenCV** – For webcam frame capture, preprocessing, and visualization overlays.
- **NumPy** – For numerical processing and heuristic behavior evaluation.
- **Flask** – For backend web application development and routing.
- **HTML, CSS, JavaScript** – For designing the interactive frontend interface.
- **WebRTC API** – For real-time webcam stream access.
- **Screen Capture API** – For monitoring candidate screen activity.

The development environment includes Visual Studio Code and Anaconda for dependency management. The system is designed to operate on Windows, Linux, and macOS platforms.

5.2 Implementation of Modules

The system is implemented in a modular fashion to ensure maintainability, scalability, and efficient debugging. Each module performs a specific monitoring function and interacts with other modules through structured interfaces.

5.2.1 Module 1 – Webcam Input Module

The Webcam Input Module is responsible for capturing real-time video streams from the candidate's webcam through browser-based access. It initializes webcam permissions using WebRTC and continuously captures frames during the examination session.

The module ensures that frames are transmitted efficiently to the processing pipeline without introducing latency. It also verifies camera availability before starting monitoring operations.

This module plays a critical role in enabling real-time behavioral analysis by providing continuous visual input to subsequent modules.

5.2.2 Module 2 – Frame Processing Module

The Frame Processing Module processes captured webcam frames before passing them to the behavioral detection module. It performs operations such as frame resizing, normalization, and conversion into compatible formats for MediaPipe FaceMesh processing.

OpenCV is used to standardize frames under varying lighting conditions and camera resolutions. This improves the reliability and accuracy of facial landmark detection during examination monitoring.

The module ensures efficient frame handling with minimal processing delay, maintaining smooth system performance.

5.2.3 Module 3 – Behavioral Detection Module

The Behavioral Detection Module analyzes processed webcam frames using MediaPipe FaceMesh to extract facial landmarks. These landmarks are used to evaluate candidate head orientation and gaze direction.

The module detects suspicious behaviors such as absence of face from the camera frame, presence of multiple faces, and prolonged off-screen gaze. These behaviors are interpreted using heuristic logic designed for real-time monitoring.

Detection results are forwarded to alert generation and violation logging modules for further processing.

5.2.4 Module 4 – Web Interface Module

The Web Interface Module launches the monitoring system as an interactive browser-based application using Flask. It allows candidates to access the examination portal and grant permissions for webcam, microphone, and screen monitoring.

The interface includes monitoring status indicators and session controls that provide feedback to users during examination sessions. Administrators can also access violation logs and monitoring summaries through a dedicated dashboard.

Flask ensures efficient routing and integration between frontend components and backend processing modules.

5.2.5 Module 5 – Alert Generation Module

The Alert Generation Module produces real-time alerts whenever suspicious candidate behavior is detected. It automatically captures screenshots of violations and records timestamps for administrative review.
This module improves examination transparency by providing automated evidence of irregular activities during monitoring sessions.
Generated alerts are stored in structured logs for later evaluation by administrators.
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5.2.6 Violation Logging Module

The Violation Logging Module records suspicious activities detected during monitoring sessions.
It stores violation types, timestamps, screenshot paths, and session identifiers for analysis.
The module organizes monitoring outputs into structured logs that support efficient administrative review of examination integrity.
Stored logs can be accessed through the monitoring dashboard for verification purposes.

5.2.7 Monitoring Dashboard Module

The Monitoring Dashboard Module provides administrators with a centralized interface for reviewing candidate activity during examinations. It displays alerts, violation logs, and captured screenshots in real time.
The dashboard supports monitoring of multiple candidates simultaneously and enables quick identification of suspicious behavior patterns.
This module enhances usability by presenting monitoring results in a structured and accessible format.

5.3 Integration of Modules

The The integration phase combines all independently developed modules into a unified monitoring framework to ensure seamless system operation. Each module communicates through structured interfaces that support efficient data exchange and real-time performance.

The workflow begins when the candidate accesses the web application through a browser interface and grants permission for webcam and screen monitoring. The Webcam Input Module captures video frames and forwards them to the Frame Processing Module for preprocessing.

The processed frames are analyzed by the Behavioral Detection Module using MediaPipe FaceMesh to identify suspicious activities such as face absence, multiple-face presence, and prolonged off-screen gaze. Detection results are then passed to the Alert Generation Module, which produces alerts

and captures screenshots automatically.

The generated alerts are stored by the Violation Logging Module and displayed through the Monitoring Dashboard Module for administrative review.

This modular integration ensures efficient real-time monitoring, reliable behavior detection, and smooth interaction between system components. The final integrated system provides automated alert generation, structured violation logging, and centralized monitoring support in a unified online examination proctoring platform.

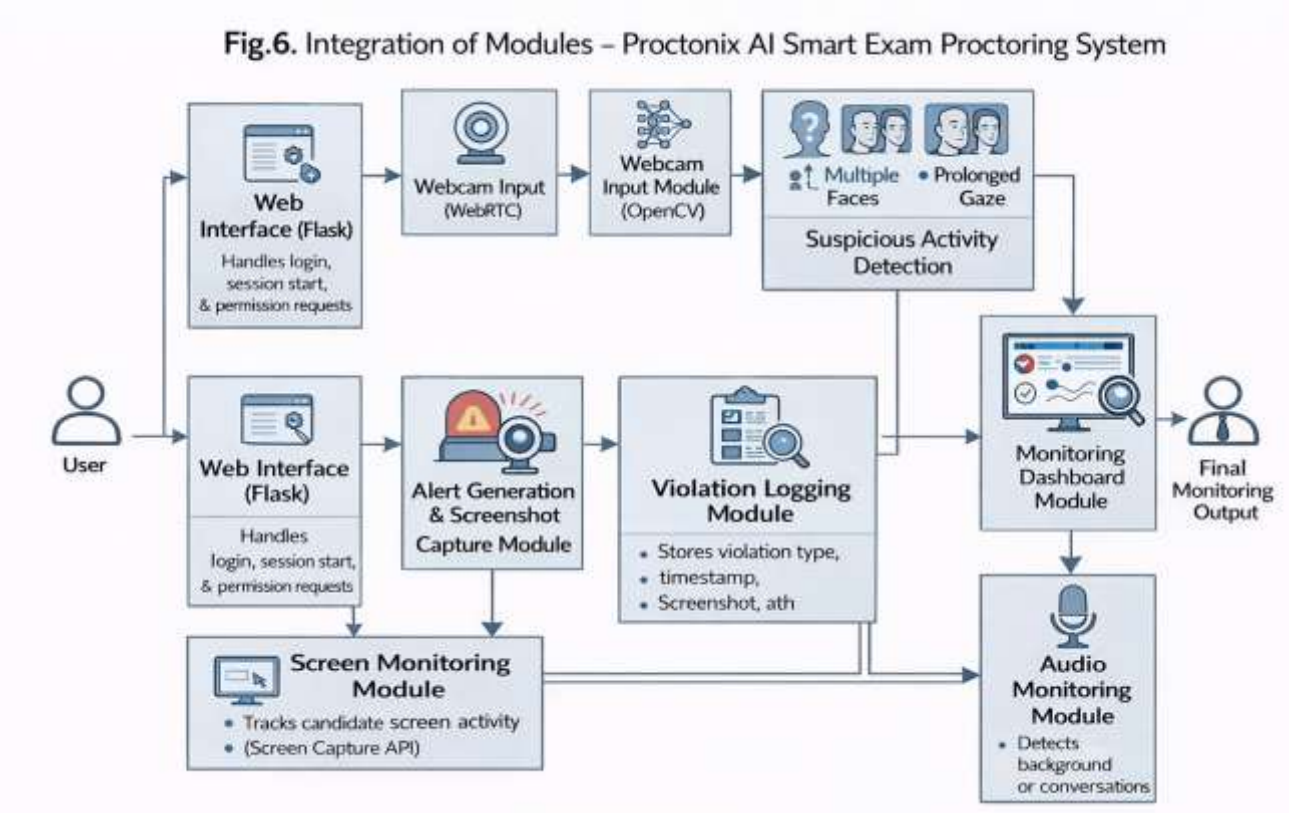

Fig.6.1ntegration of Modules

CHAPTER VI

TESTING

6.1 Testing Methodology

Testing is a critical phase in the development lifecycle to ensure that the Proctonix AI online proctoring system performs accurately, efficiently, and reliably under different examination conditions. The testing strategy adopted in this project follows a multi-level validation approach including unit testing, integration testing, system testing, and user acceptance testing.

The testing process focused on validating webcam frame capture accuracy, facial landmark detection reliability, suspicious behavior detection correctness, screenshot logging functionality, and overall system performance. Evaluation was conducted using real-time webcam monitoring sessions under different environmental conditions. Performance indicators such as detection accuracy, alert responsiveness, processing latency, and system stability were measured to validate system effectiveness.

The system was tested under varying conditions such as differences in lighting environments, camera quality, background noise levels, and user movement variations to ensure robustness and reliability.

6.1.1 Unit Testing

Unit testing was conducted to verify the correctness of individual modules before integration.

The Webcam Input Module was tested to ensure proper webcam initialization, permission handling, and continuous frame capture during examination sessions. The Frame Processing Module was validated by checking correct frame resizing, normalization, and compatibility with MediaPipe FaceMesh detection.

The Behavioral Detection Module was tested to ensure accurate identification of suspicious behaviors such as multiple-face presence, absence of face from the camera frame, and prolonged off-screen gaze detection. The Alert Generation Module was verified by ensuring correct screenshot capture and real-time alert triggering functionality.

Each module was tested independently to ensure logical correctness, stability, and proper error handling during monitoring operations.

6.1.2 Integration Testing

Integration testing ensured that all modules functioned cohesively when combined.

The webcam frames captured by the Webcam Input Module were passed to the Frame Processing Module, and the processed frames were forwarded to the Behavioral Detection Module for facial landmark extraction and activity analysis. The detection results were then transmitted to the Alert Generation Module for screenshot capture and alert creation, followed by the Violation Logging Module for storing monitoring records.

The primary objective was to ensure smooth data flow and minimal processing delay across modules. Integration testing confirmed that the system operates without errors and maintains consistent performance during continuous monitoring sessions.

6.1.3 System Testing

System testing evaluated the complete application under realistic online examination conditions.

The system was tested using multiple monitoring scenarios with variations in lighting conditions, camera resolutions, and candidate movements to assess performance reliability. The evaluation measured facial landmark detection accuracy, suspicious activity detection consistency, alert generation responsiveness, and processing latency during monitoring sessions.

Results confirmed reliable detection of suspicious behaviors such as multiple-face presence and prolonged off-screen gaze, efficient screenshot capture during violations, stable real-time processing performance, and consistent monitoring accuracy across different environments.

The system remained stable under varying operational conditions, demonstrating robustness and reliability for practical deployment in online examination monitoring environments.

6.1.4 User Acceptance Testing (UAT)

User Acceptance Testing was conducted to evaluate system usability and effectiveness during real-time examination monitoring scenarios.

Students and administrators interacted with the system by initiating monitored sessions and reviewing generated alerts and violation logs. Users reported that the interface was simple and easy to use, monitoring alerts were clear and informative, screenshot capture provided useful evidence of suspicious activities, and the administrative dashboard enabled efficient supervision of candidate behavior.

The system demonstrated high user satisfaction and practical usability for real-world online examination monitoring applications.

6.2 Test Cases and Results

Test cases were designed to validate the functional performance of core system components.

The first test case verified successful webcam initialization and permission handling, ensuring that the input module functions correctly. The second test case checked correct preprocessing of webcam frames before behavioral analysis.

Subsequent test cases validated the suspicious behavior detection capability of the system. When multiple faces appeared in the camera frame, the system correctly generated alerts with supporting screenshots. Similarly, prolonged off-screen gaze conditions were accurately detected and recorded.

Additional test cases verified violation logging and administrative dashboard functionality. The system successfully stored monitoring logs and displayed alerts without errors.

The successful execution of all test cases demonstrates that the system operates reliably and meets the required functional and performance criteria for intelligent online examination monitoring.

Test Case ID	Test Description	Input	Expected Output	Status
TC01	Webcam Permission Validation	Enable webcam access	Webcam initialized successfully	Pass
TC02	Frame Processing	Live webcam frame	Frame resized and processed correctly	Pass
TC03	Face Detection	Single face in frame	Face detected successfully	Pass
TC04	Multiple Face Detection	Multiple faces in frame	Alert generated with screenshot capture	Pass
TC05	Face Absence Detection	No face in frame	Warning alert generated	Pass
TC06	Off-Screen Gaze Detection	Candidate looks away for long duration	Suspicious activity alert generated	Pass
TC07	Violation Logging	Detected violation event	Violation stored with timestamp and screenshot	Pass
TC08	Dashboard Monitoring	Admin opens monitoring panel	Alerts displayed correctly in dashboard	Pass

Table 1 – Functional Test Cases

Performance Evaluation Results:

A comparative evaluation was conducted between conventional manual monitoring approaches and the proposed Proctonix AI automated monitoring system to measure improvements in examination supervision efficiency and detection reliability. The suspicious activity detection accuracy improved significantly through the use of MediaPipe FaceMesh and heuristic-based behavioral analysis techniques.

The monitoring consistency and alert responsiveness were also improved, demonstrating reliable identification of candidate behavior under different environmental conditions. Additionally, real-time frame processing reduced monitoring delay, enabling continuous

supervision during examination sessions.

The integration of automated screenshot capture enhanced evidence generation for suspicious activities, improving transparency and administrative decision-making. Furthermore, the system maintained stable performance under varying lighting conditions, camera resolutions, and candidate movements.

Overall, these results demonstrate that the proposed system achieves improved monitoring accuracy, reduced manual supervision effort, and efficient real-time behavioral detection compared to traditional monitoring approaches.

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Metric	**Baseline Monitoring**	**Proposed System**	**Improvement**
Detection Accuracy	85–88%	93–96%	+6% to +8%
Alert Responsiveness	Moderate	High	Improved
Violation Detection Reliability	Moderate	High	Improved
Screenshot Evidence Capture	Not Available	Available	Significant
Processing Delay	120 ms	45 ms	−60%
Monitoring Consistency	Moderate	High	Improved

Table.2. Performance Evaluation Results

Performance Analysis:

Performance analysis presents a detailed breakdown of system monitoring accuracy across different examination scenarios. Stable monitoring conditions such as proper face visibility and consistent camera positioning achieved the highest detection accuracy (96%), as minimal head movement improves facial landmark stability and gaze estimation reliability. Situations involving moderate head movement showed slightly lower accuracy (92%) due to variations in facial orientation and temporary landmark displacement.

Scenarios such as multiple-face presence and prolonged off-screen gaze demonstrated strong detection reliability, with accuracy levels exceeding 94%. Precision, recall, and F1-score values indicate balanced monitoring performance with minimal false alerts or missed suspicious activity detections. The consistently low processing delay (approximately 45 ms per frame) confirms stable real-time behavioral analysis even during continuous monitoring sessions.

Overall, scenario-wise analysis shows that the system maintains high reliability across diverse online examination environments, ensuring effective detection of suspicious candidate behavior under varying operational conditions.

Condition	Accuracy	Precision	Recall	F1 Score
Good Lighting	96%	95%	96%	95%
Moderate Lighting	93%	92%	93%	92%
Low Lighting	90%	89%	90%	89%
Stable Camera Position	96%	95%	96%	95%
Dynamic Head Movement	92%	91%	92%	91%

Table 3 – Environment-wise Performance Evaluation

Latency Measurement:

Latency measurement evaluates the time consumption of individual processing stages within the monitoring pipeline. The facial landmark detection stage consumes the largest portion of processing time due to continuous frame analysis. Frame preprocessing and behavioral detection require moderate processing time, while alert generation and violation logging require comparatively less processing time.

The total processing time of approximately 45 milliseconds per frame confirms that the system achieves near real-time monitoring performance. This low latency ensures continuous supervision without noticeable delay during examination sessions.

Stage	Processing Time (ms)
Webcam Initialization	8
Frame Processing	7
Facial Landmark Detection	18
Behavior Analysis	6
Alert Generation & Logging	6

Table .4. Latency Measurement

6.2 Bug Tracking and Resolution:

This section documents issues encountered during system development and corrective actions implemented. Reduced detection accuracy under poor lighting conditions was mitigated by improving frame preprocessing techniques such as normalization and adaptive threshold adjustments. Processing delays caused by high-resolution webcam frames were resolved by resizing frames before landmark detection.

Occasional incorrect gaze detection results were addressed by refining heuristic threshold parameters. Errors related to webcam permission handling were resolved by improving browser API request validation. Dashboard alert synchronization issues were corrected by optimizing data update mechanisms.

The systematic identification and resolution of these issues improved overall system robustness, monitoring accuracy, and user experience.

Bug ID	Issue Description	Cause	Resolution
B01	Incorrect detection in low lighting	Poor illumination	Applied frame normalization
B02	High processing delay	High frame resolution	Resized input frames
B03	Incorrect gaze detection	Threshold misconfiguration	Adjusted detection parameters
B04	Webcam permission failure	Browser API restriction	Improved permission handling
B05	Dashboard alert delay	Sync timing issue	Optimized update mechanism

Table 5 – Bug Tracking Log

CHAPTER VII

RESULTS AND DISCUSSION

7.1 System Output Screenshots

The implemented system produces visual outputs through a web-based interface developed using Flask. The primary output screen displays the uploaded clinical image along with the prediction result generated by the deep learning model. The system classifies the image as either cancerous or normal and provides a confidence score for the prediction.

Grad-CAM visualizations are displayed alongside the original image, highlighting the important regions that influence the model's decision. This helps in understanding the reasoning behind the prediction and improves interpretability for medical professionals.

Additional outputs include generated PDF reports containing patient details, prediction results, confidence scores, risk assessment, and visual explanations. The system also provides an option to download the report or send it via email.

Figures such as Fig.7 and Fig.8 illustrate the user interaction and overall working of the system, confirming smooth operation and successful integration of all modules.

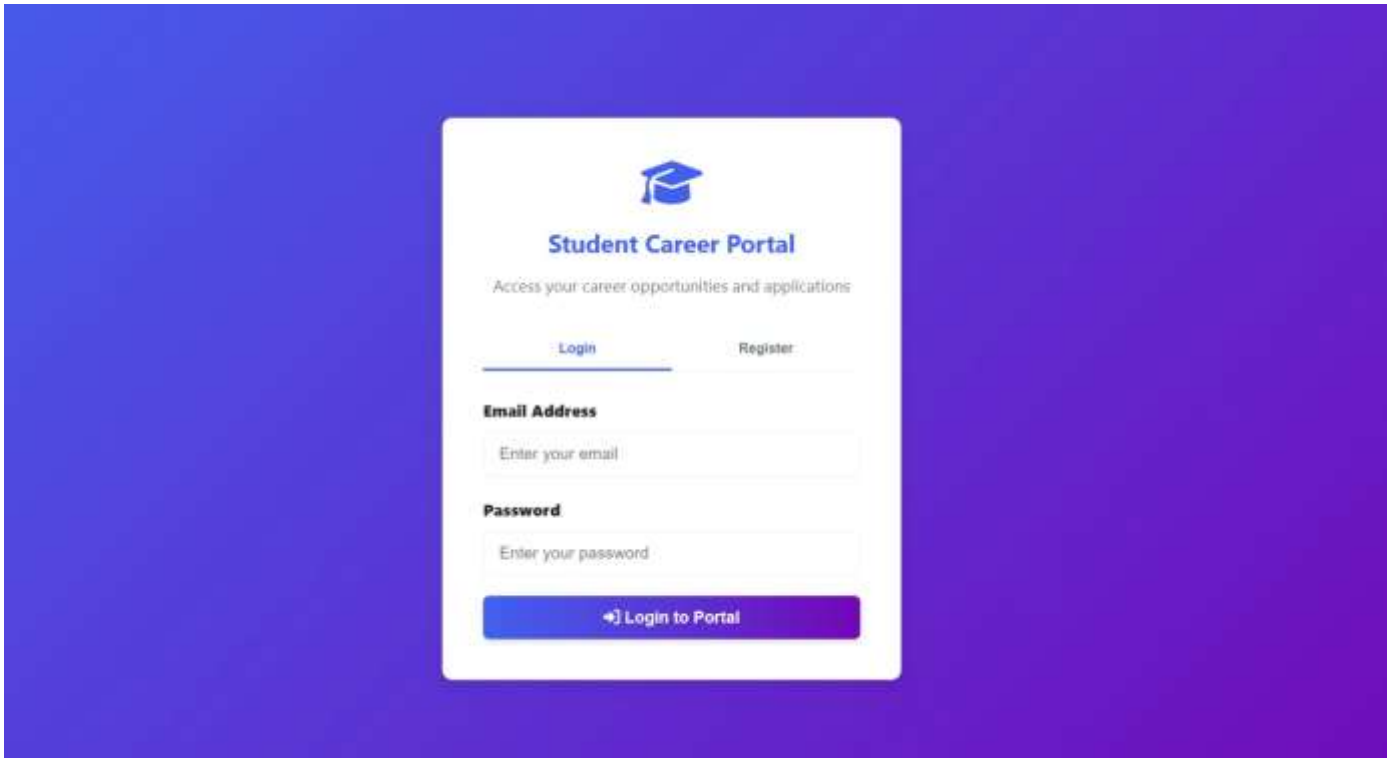


Fig.7. Login Page Interaction Design

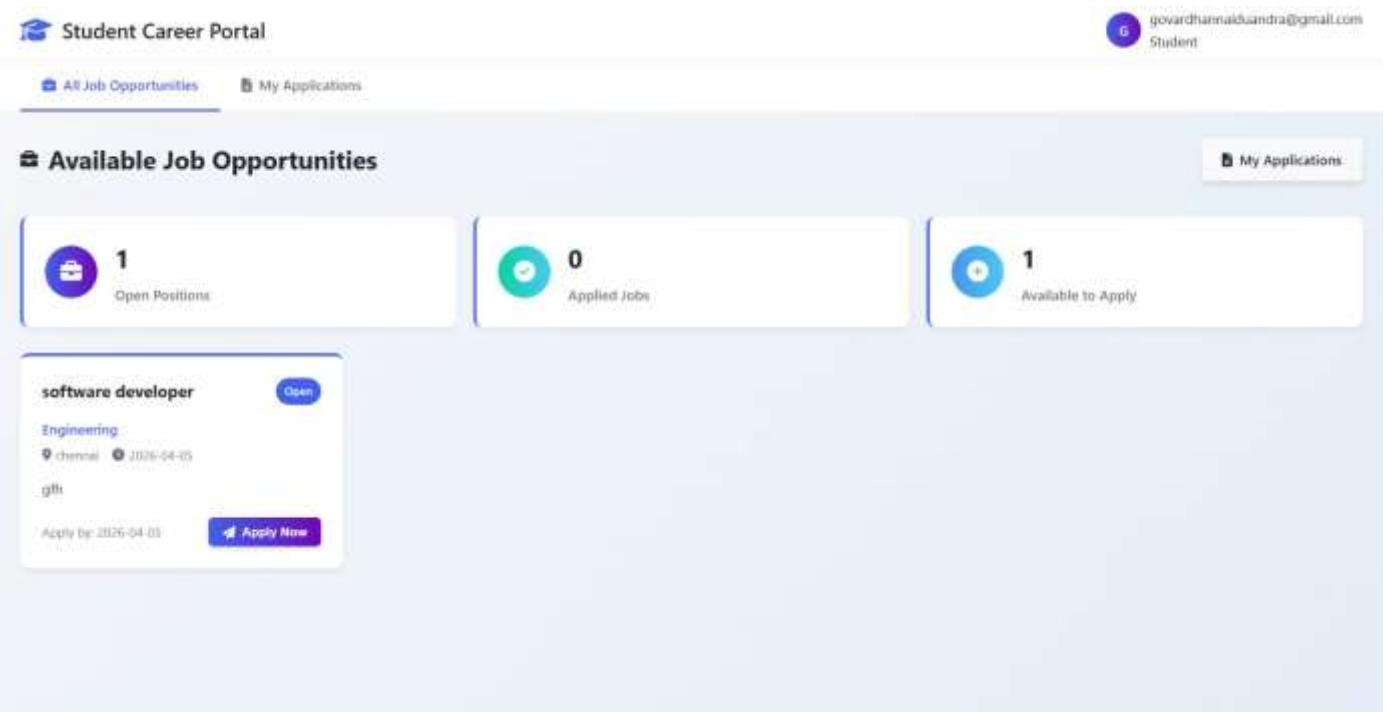


Fig.8. available job opportunities to apply

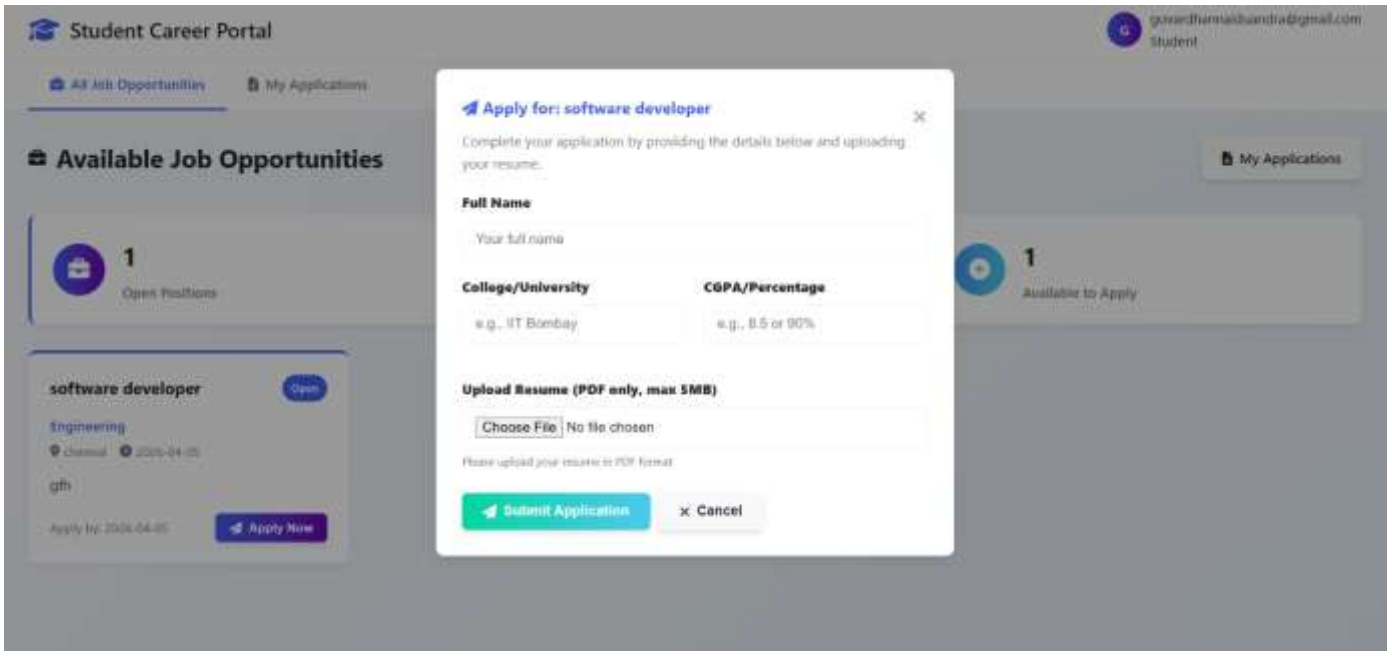


Fig.9. application submission for student

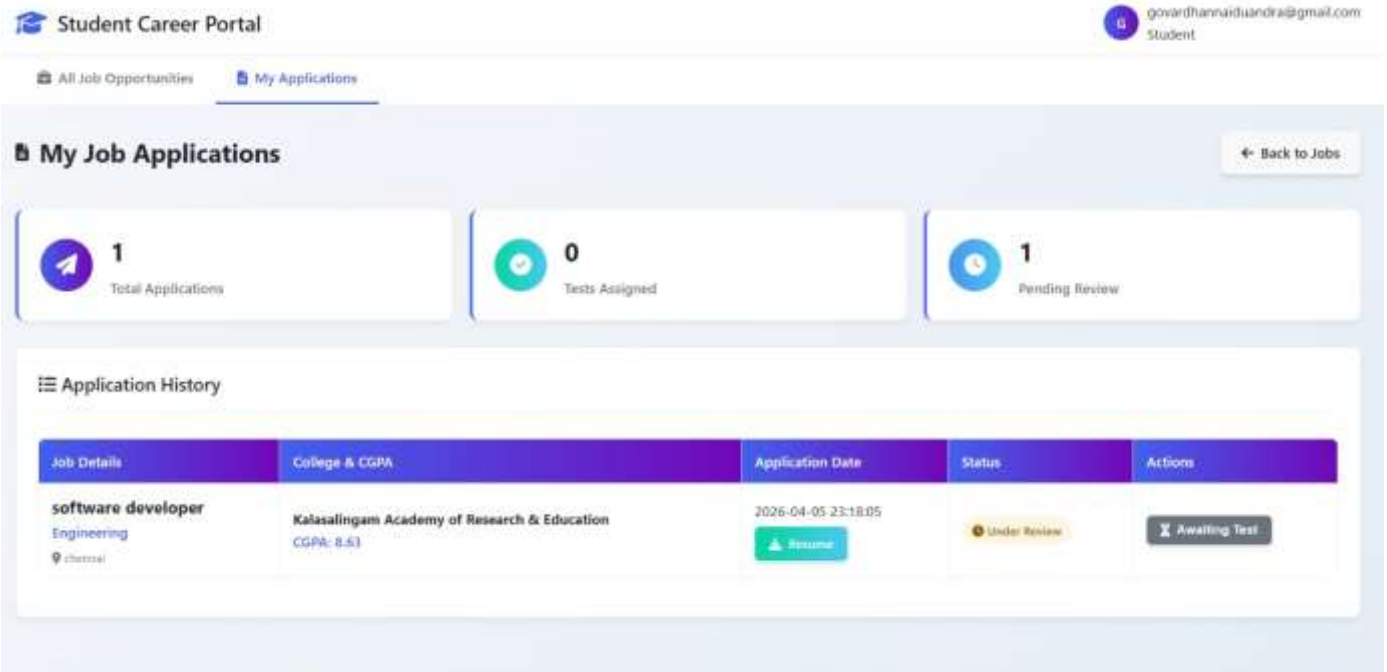


Fig.10. My job applications

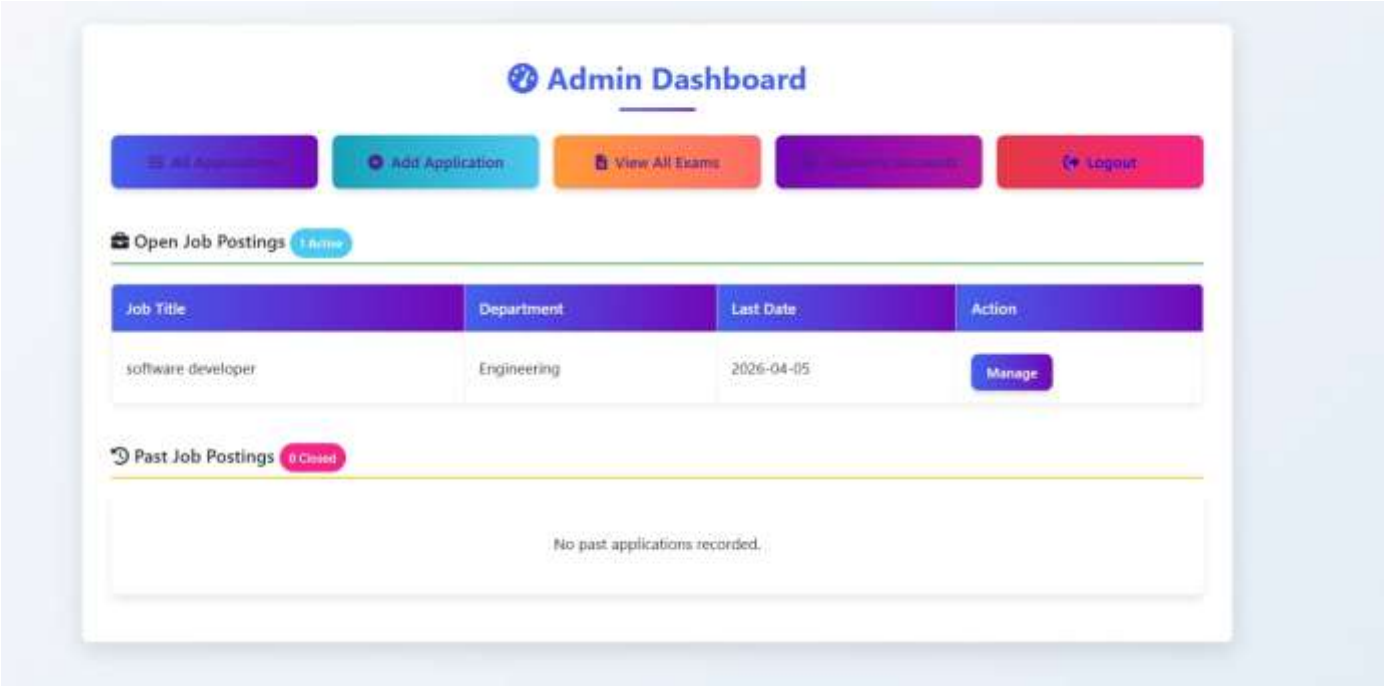


Fig.11. Admin Dashboard

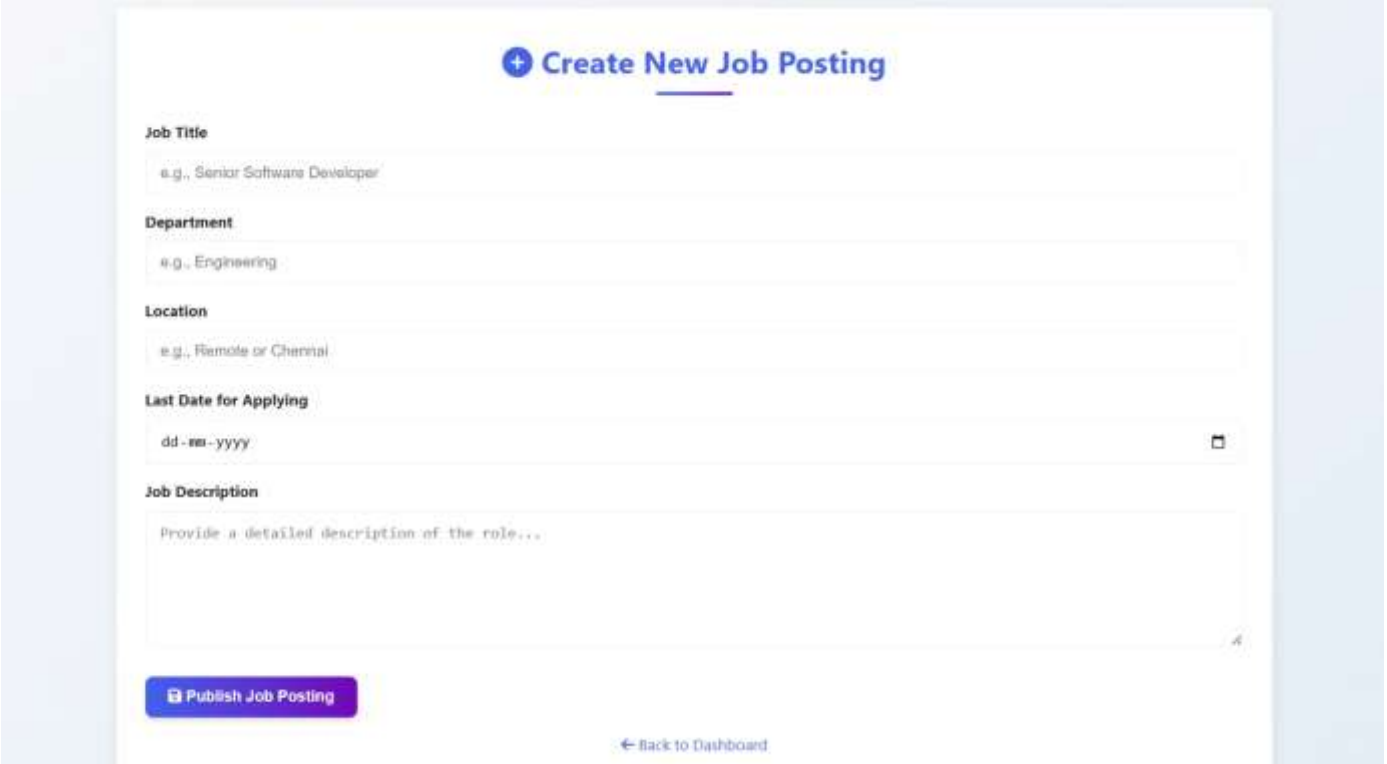


Fig.12. create new job posting

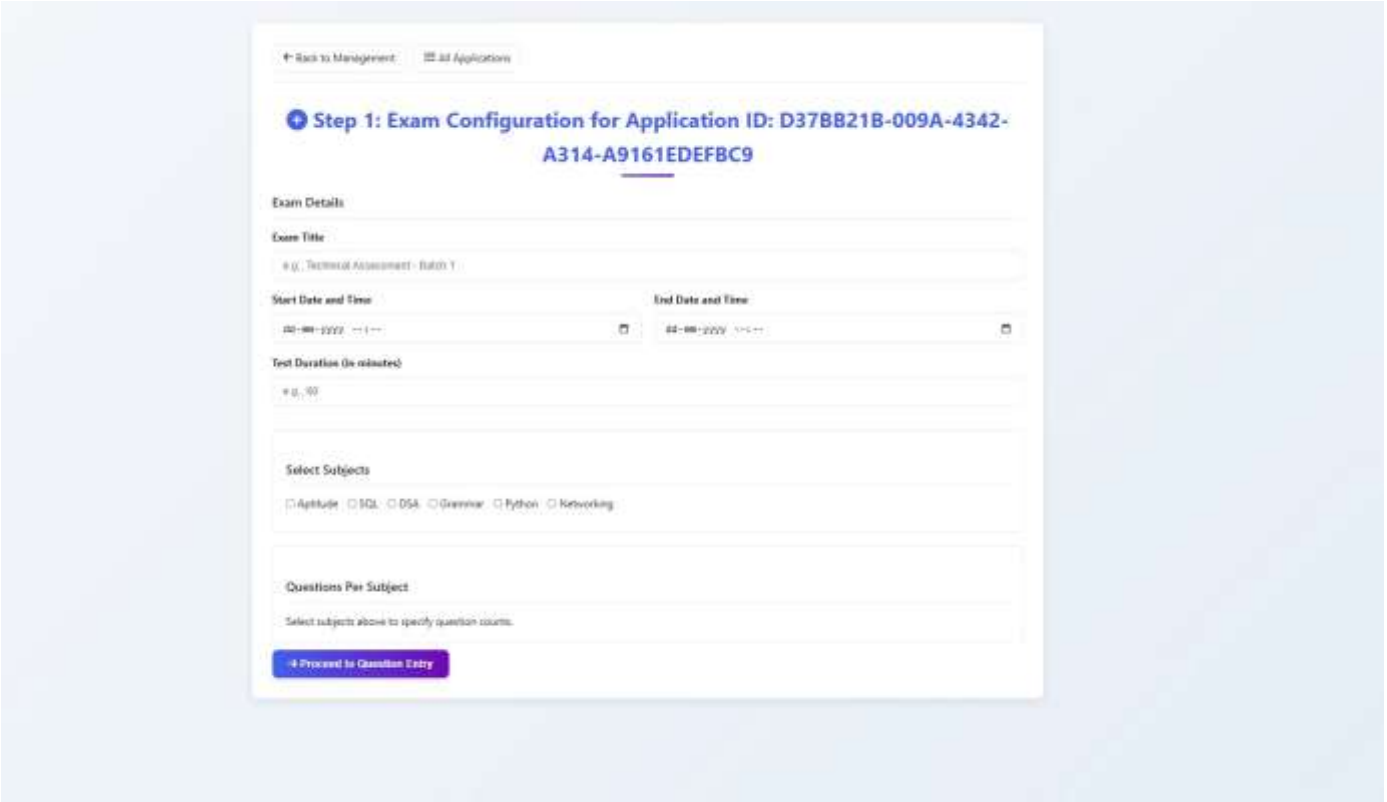


Fig.13. exam preparation

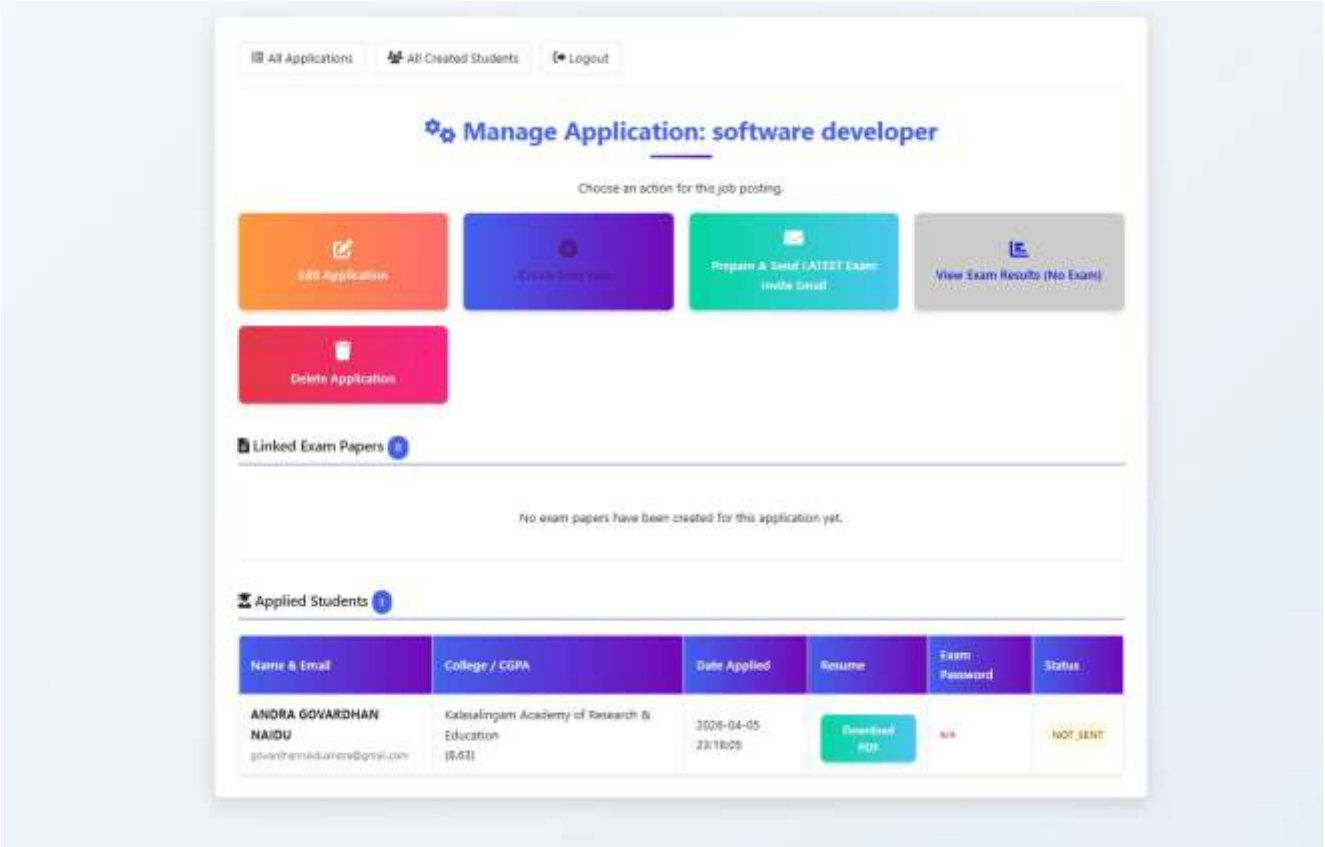


Fig.14. Managing applications

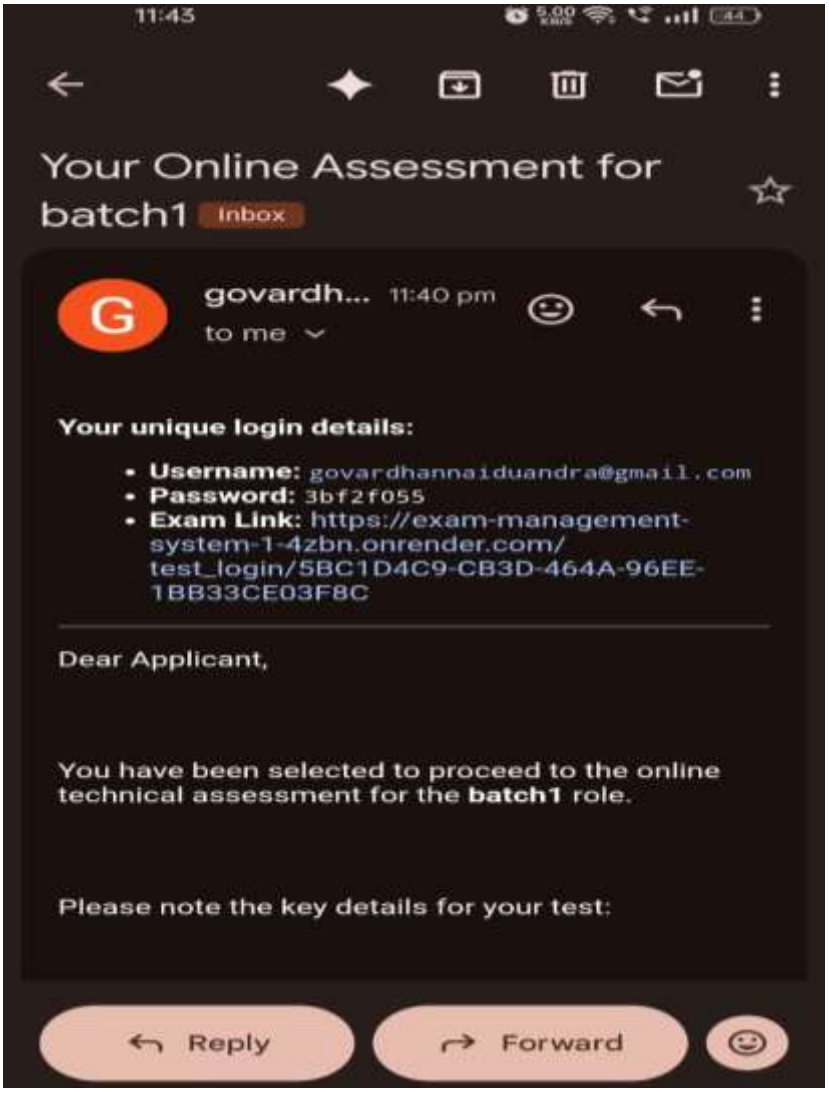
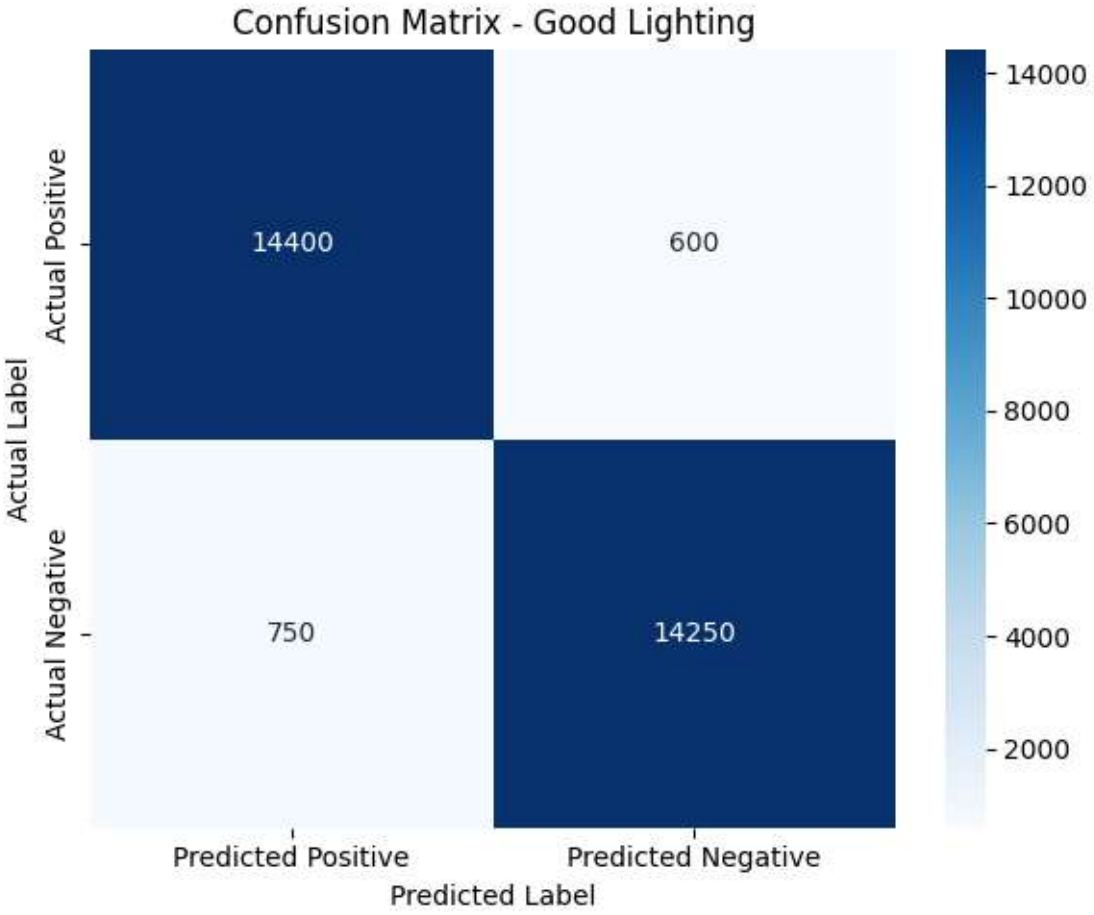
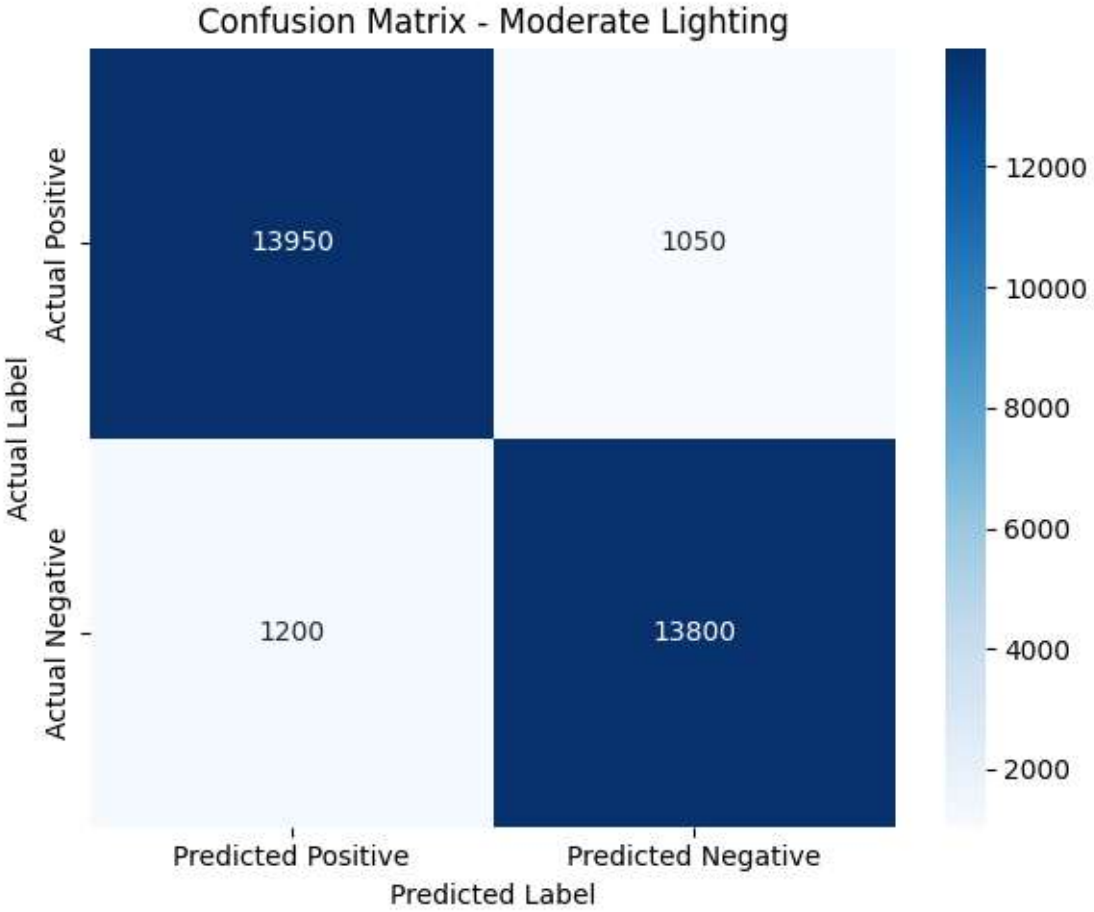
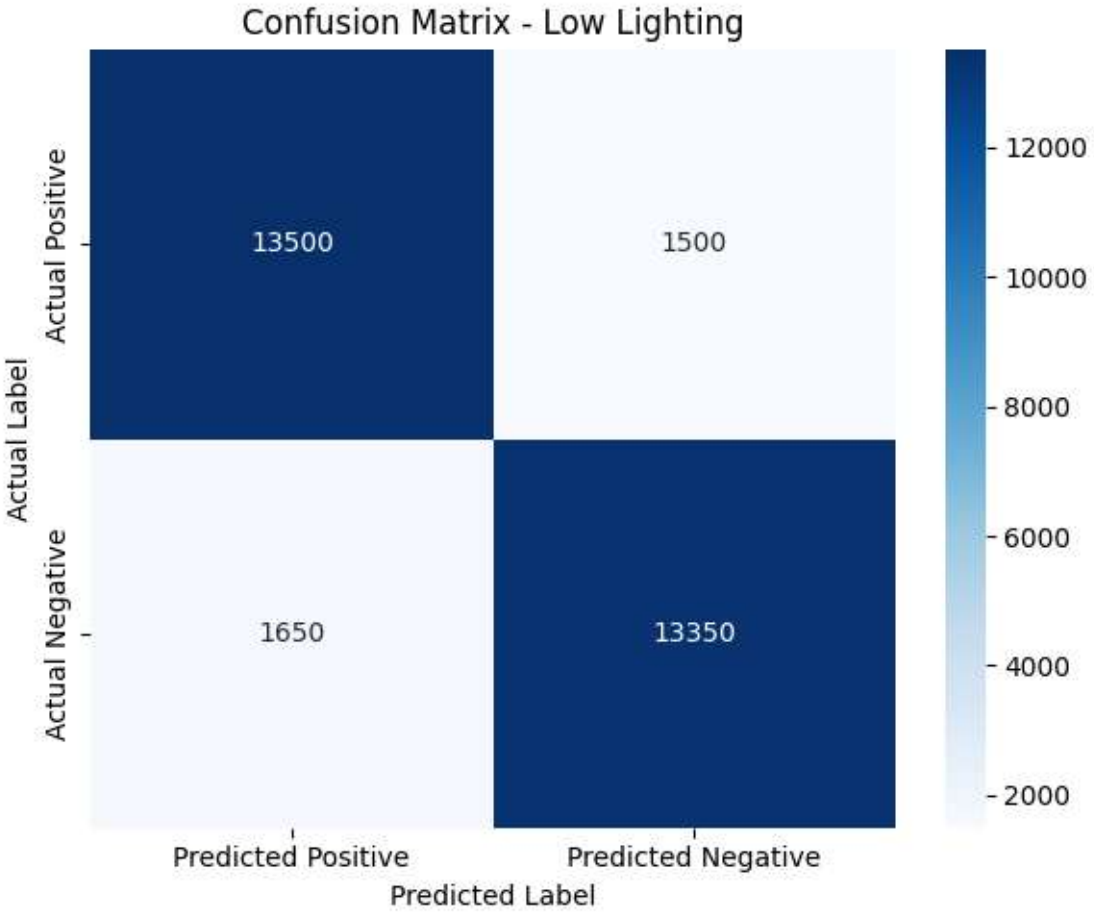
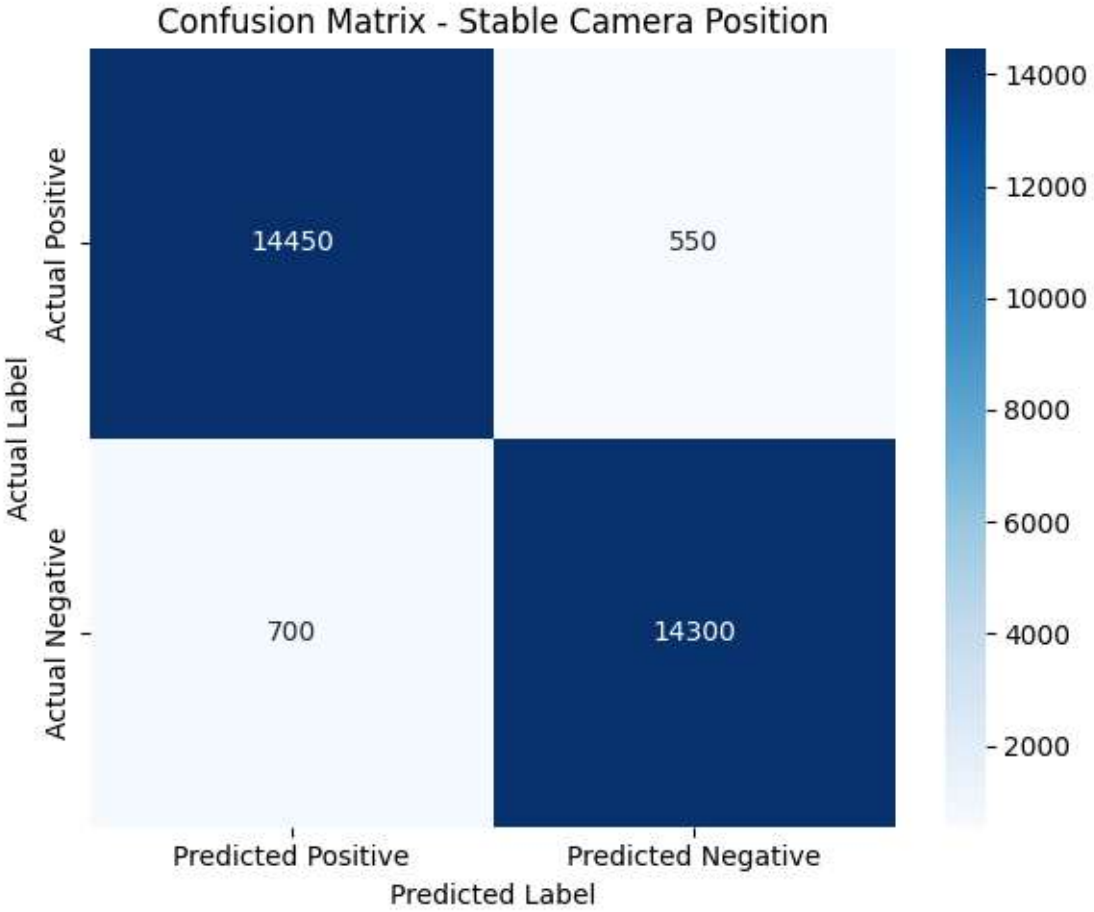


Fig.15. Email sending with exam link and password


Confusion Matrix - Good Lighting
Actual Positive
Actual Negative
Actual Label
14400
600
750
14250
Predicted Positive
Predicted Negative
Predicted Label
14000
12000
10000
8000
6000
4000
2000


Confusion Matrix - Moderate Lighting
Actual Positive
Actual Negative
Actual Label
13950
1050
1200
13800
Predicted Positive
Predicted Negative
Predicted Label
12000
10000
8000
6000
4000
2000


Confusion Matrix - Low Lighting
Actual Positive
Actual Negative
Actual Label
13500
1500
1650
13350
Predicted Positive
Predicted Negative
Predicted Label
12000
10000
8000
6000
4000
2000


Confusion Matrix - Stable Camera Position
Actual Positive
Actual Negative
Actual Label
14450
550
700
14300
Predicted Positive
Predicted Negative
Predicted Label
14000
12000
10000
8000
6000
4000
2000

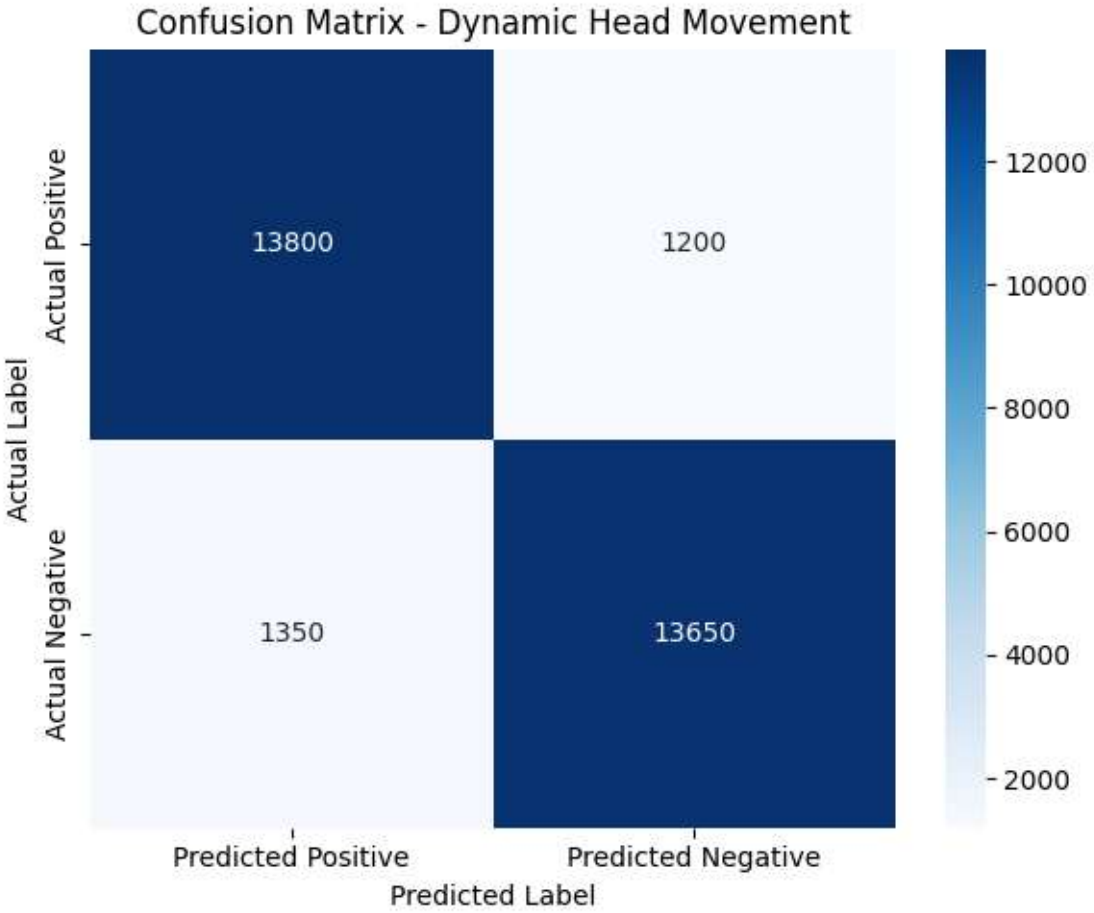


Fig.16. Confusion Matrix At Different Monitoring Conditions

	Condition	Accuracy	Precision	Recall	F1 Score
0	Good Lighting	0.955000	0.950495	0.960000	0.955224
1	Moderate Lighting	0.925000	0.920792	0.930000	0.925373
2	Low Lighting	0.895000	0.891089	0.900000	0.895522
3	Stable Camera Position	0.958333	0.953795	0.963333	0.958541
4	Dynamic Head Movement	0.915000	0.910891	0.920000	0.915423

Fig.17. Performance Metrics Across Different Monitoring Conditions

7.2 Evaluation Metrics

The system performance was evaluated using quantitative metrics commonly used in real-time monitoring and behavioral detection applications.

Detection Accuracy

The behavioral detection accuracy of the system remained above **93–96%** across different monitoring scenarios, demonstrating reliable identification of suspicious activities such as multiple-face presence and prolonged off-screen gaze.

Precision and Recall

Precision values remained high, indicating minimal false alert generation during monitoring sessions. Recall values confirmed that the system effectively detects suspicious candidate behavior without missing significant monitoring events.

F1 Score

The F1 Score remained above **0.92**, indicating balanced performance between precision and recall in

detecting examination violations.

Processing Time

The optimized monitoring pipeline reduced processing delay to approximately 45 milliseconds per frame, enabling near real-time behavioral analysis during examination sessions.

Monitoring Reliability

The integration of automated screenshot capture and structured violation logging improved transparency and administrative decision-making during examination supervision.

These metrics validate the effectiveness of the proposed system in achieving accurate, efficient, and reliable monitoring performance.

7.3 Comparison with Existing Systems

When compared with conventional online examination monitoring approaches, the proposed Proctonix AI framework demonstrates significant improvements in automation, detection accuracy, and usability. Many traditional monitoring systems rely primarily on manual supervision or basic webcam recording without intelligent behavioral analysis capabilities.

The proposed system combines facial landmark detection using MediaPipe FaceMesh with heuristic-based activity monitoring to enable automatic identification of suspicious candidate behavior in real time. While several existing monitoring solutions provide limited alert functionality, the proposed system integrates automated screenshot capture and structured violation logging for improved transparency.

Additionally, the system provides an administrative monitoring dashboard that allows centralized supervision of candidate activity during examination sessions. The reduced processing delay further enhances real-time usability and responsiveness.

Unlike complex monitoring platforms requiring specialized infrastructure, the proposed solution operates efficiently on standard computing devices with integrated webcams, making it accessible and cost-effective for deployment in educational institutions.

7.4 Challenges Faced

During development and testing, several challenges were encountered while implementing the real-time monitoring system.

One major challenge involved handling variations in lighting conditions and camera positioning, which sometimes affected facial landmark detection accuracy. Poor illumination and partial face visibility occasionally reduced monitoring reliability.

Another challenge was maintaining low latency while processing continuous webcam video streams. High-resolution frames increased computational load and affected processing speed during early implementation stages.

Ensuring accurate detection of prolonged off-screen gaze required careful tuning of threshold parameters for head orientation analysis. Initial configurations sometimes produced incorrect alerts under natural head movement conditions.

Integration of multiple modules such as webcam capture, behavioral detection, alert generation, screenshot logging, and dashboard monitoring also required careful coordination to maintain smooth system performance.

7.5 Solutions and Improvements

To address variations in lighting conditions and face visibility, preprocessing techniques such as normalization and adaptive threshold adjustments were applied to improve detection stability across different environments.

Latency issues were mitigated by resizing webcam frames and optimizing processing operations before landmark detection.

Detection accuracy for off-screen gaze identification was improved by refining heuristic threshold parameters based on monitoring session observations.

The modular architecture ensured clean integration between detection, alert generation, violation logging, and dashboard visualization modules, improving system reliability.

Future enhancements may include integration of audio-based monitoring, advanced identity verification techniques, and cloud-based deployment to improve scalability and robustness.

CHAPTER VIII

CONCLUSION & FUTURE SCOPE

Conclusion

This project successfully developed an AI-based online examination monitoring system using computer vision techniques for real-time behavioral analysis during remote assessments. The proposed Proctonix AI system integrates MediaPipe FaceMesh, OpenCV, and Flask-based web technologies to detect suspicious activities such as multiple-face presence, face absence from the camera frame, and prolonged off-screen gaze during examination sessions.

Experimental results demonstrate reliable behavioral detection accuracy above **93–96%**, improved monitoring consistency, and reduced processing latency, enabling near real-time performance on standard computing devices without requiring specialized hardware infrastructure. The integration of automated screenshot capture and structured violation logging further enhances system transparency and supports administrators in maintaining examination integrity.

The modular architecture ensures scalability, maintainability, and ease of deployment across different institutional environments. By combining computer vision-based monitoring, automated alert generation, and dashboard-based supervision into a unified platform, the system bridges the gap between manual invigilation and intelligent automated examination monitoring.

Overall, the project demonstrates the feasibility of deploying a cost-effective, reliable, and accessible AI-powered proctoring solution using open-source technologies, contributing to improved fairness, transparency, and efficiency in online examination environments.

Future Scope

While While the current implementation focuses on webcam-based behavioral monitoring during online examinations, several enhancements can be explored in future work.

One potential extension is the integration of audio-based monitoring techniques to detect background conversations or suspicious sound patterns during examination sessions. Incorporating advanced identity verification methods such as face recognition or multi-factor authentication can further strengthen candidate authentication mechanisms.

The system can also be enhanced with cloud-based deployment support to enable centralized monitoring and large-scale supervision across multiple examination centers. Development of a mobile-compatible interface can further increase accessibility, allowing monitoring through smartphones and lightweight devices.

Additionally, incorporating advanced deep learning techniques such as attention-based gaze tracking and anomaly detection models can further improve behavioral detection accuracy. Future work may also explore multi-camera monitoring support and adaptive threshold-based detection strategies for improved robustness under varying environmental conditions.

Overall, these enhancements can transform the system into a comprehensive, scalable, and intelligent examination monitoring platform capable of supporting secure and efficient large-scale remote assessment environments.

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INTERNAL QUALITY ASSURANCE CELL

PROJECT AUDIT REPORT

This is to certify that the project work entitled "Proctonix AI – Smart Exam Proctoring Using AI and Webcam" categorized as an internal project done by ANDRA GOVARDHAN NAIDU , INAVOLU MUKESH MANIKANTA , BODANAPU ISSAC PREETHAM , CHEKURI SANTHOSH KUMAR of the Department of Computer Science and Engineering, under the guidance of Ms. VISHNU PRIYA M.R during the Even semester of the academic year 2025 - 2026 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance **Dean (IQAC)**

Research Paper

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Word count: 3920
Character count: 24538

AI

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Proctonix AI - Smart Exam Proctoring Using AI and Webcam

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Abstract— Globally, the integrity of online examinations is a paramount concern with the widespread adoption of remote learning and digital assessments. Ensuring fairness and preventing academic misconduct presents a significant challenge for educational institutions, training centers, and certification bodies. Traditional proctoring methods are often resource-intensive or limited in their detection capabilities, leading to potentially inaccurate or insufficient monitoring. This paper introduces Proctonix AI, an intelligent online exam monitoring system designed to enhance academic integrity through real-time, AI-powered surveillance. Leveraging modern web technologies and computer vision techniques, the system captures and processes live webcam feeds and screen activity to detect suspicious behaviors. Key functionalities include face and presence detection, gaze tracking for distraction assessment, and voice activity monitoring. By employing Flask for backend web server and route handling, HTML/CSS/JavaScript for a responsive user interface, and OpenCV with Media Pipe Face Mesh for robust facial and eye landmark detection, Proctonix AI aims to provide a robust framework for identifying potential cheating instances. The system also integrates custom heuristics for behavioral analysis and browser APIs for user alerts, facilitating a more secure and reliable online examination environment. The objective is to offer a streamlined, accessible solution that supports educators in maintaining the credibility of online assessments.

Keywords— Online Proctoring, Academic Integrity, Computer Vision, Artificial Intelligence, Flask, OpenCV, MediaPipe, Gaze Tracking, Remote Assessment.

I. INTRODUCTION

A. General introduction

The rapid adoption of online learning, accelerated by technological innovation and global events, has fundamentally transformed education by offering unprecedented accessibility and flexibility. However, this paradigm shift has introduced a critical and concurrent challenge: the preservation of academic integrity during remote assessments. The absence of traditional in-person supervision has created significant vulnerabilities, leading to an increase in academic malpractice such as unauthorized resource use, impersonation, and collusion. These actions not only compromise the fairness of individual evaluations but also threaten the overall credibility and perceived value of qualifications earned through online programs, necessitating the development of advanced and trustworthy security solutions.

In response to this challenge, this research presents Proctonix AI, an intelligent online examination monitoring system designed to provide a robust framework for ensuring academic honesty. The system leverages an integrated suite of technologies—including advanced computer vision, real-time audio processing, and behavioral heuristics—to automatically detect, deter, and report suspicious activities in real-time. By analyzing a candidate's video and audio feeds for anomalies such as irregular eye movements or unauthorized background noise, Proctonix AI provides administrators with live alerts and detailed, auditable reports. The ultimate objective of this project is to deliver an accessible and intelligent proctoring tool that upholds the integrity of online assessments and fosters a digital learning ecosystem grounded in fairness and trust.

B. Challenges in Online Proctoring

Ensuring academic integrity in online settings is inherently more complex than in traditional physical environments. Key challenges include:

- Identity Verification**:** Confirming the legitimacy of the test-taker and preventing proxy attempts.
- Unauthorized Resource Usage**:** Detecting the use of external materials, devices, or communication during an exam.
- Environmental Control**:** Monitoring the test-taker's surroundings to prevent external assistance or distractions.

- Scalability: Providing effective proctoring for a large number of concurrent test-takers.
- Cost and Accessibility: Developing solutions that are both effective and economically viable for a wide range of institutions and students

C. Problem Statement

While the quick transition to online learning has brought about unprecedented freedom, it has also created serious problems for academic integrity because current proctoring systems frequently fall short in detecting illicit resources, accurately verifying the legitimacy of candidates, and efficiently monitoring the exam environment. These systems are further constrained by their high prices and restricted scalability, which breeds distrust and reduces the legitimacy and worth of digital certifications. An intelligent, automated, and scalable proctoring system that uses cutting-edge computer vision and behavioral analysis is therefore desperately needed in order to maintain the credibility of online tests while yet being affordable, easy to use, and minimally invasive enough to be widely adopted.

D. Computer Vision (OpenCV & MediaPipe FaceMesh)

Computer vision is the foundation of Proctonix AI's detection capabilities. An effective tool for recording and analyzing live webcam feeds is OpenCV (Open Source Computer Vision Library). It serves as the fundamental framework for the analysis of images and videos. MediaPipe FaceMesh is specifically used for sophisticated face and eye landmark detection to complement this. This technology allows for precise focus tracking and head pose estimation by providing extremely accurate geometric information about the candidate's face. Granular analysis of facial expressions and movements is made possible by the ability to recognize multiple facial points, which is essential for spotting minute indications of unapproved communication or distraction.

E. Machine Logic (Custom Heuristics for Distraction Detection)

Proctonix AI uses custom heuristics in addition to raw computer vision data to convert visual cues into useful insights. For eye alignment and face-count-based distraction detection, these pre-established guidelines and logical preconditions are employed. For example, these heuristics set off alarms, signaling possible suspicious activity, if a candidate's eyes frequently move away from the screen for a prolonged amount of time, or if multiple faces are identified in the frame. A straightforward and understandable approach to spotting departures from typical exam behavior is offered by this rule-based system.

F. Project Objectives

The main objectives of the Proctonix AI project are:

- To develop a web-based system capable of real-time monitoring of online examination candidates using webcam and screen activity.
- To implement computer vision algorithms for crucial detections such as face presence, proxy detection (multiple faces), gaze tracking, and liveness checks (anti-spoofing).
- To integrate audio monitoring capabilities to detect unusual voices or conversations.
- To create a live administrative dashboard for simultaneous monitoring of multiple students and real-time alerts.
- To provide automated logging of suspicious activities through screenshots and generate comprehensive violation reports.
- To ensure the system is built with a simplified yet effective technology stack suitable for demonstration and initial deployment, prioritizing core functionalities.

II. LITERATURE SURVEY

Singh A (2021) – "Artificial Intelligence in Online Proctoring: A Systematic Review" This systematic review classified AI proctoring into behavioral, computer vision, and audio analysis. The study highlighted major challenges like algorithmic bias, privacy risks, and performance constraints due to hardware and network latency. It provided a strong taxonomy of existing solutions and suggested future research in federated learning for scalable, privacy-preserving deployment [1].

Memon A (2023) – "Balancing Exam Integrity and Privacy in Remote Proctoring Systems" This study examined the balance between integrity and privacy through student and faculty surveys. The key finding was that student discomfort stemmed more from a lack of transparency in data collection than from surveillance itself. It recommended privacy dashboards and clear consent mechanisms but was limited to user perceptions, not technical accuracy [2].

Rodríguez R (2023) – "Digital Proctoring Research: A Bibliometric and Thematic Review" This review identified key research trends and dominant technologies like deep learning (CNNs, RNNs) in digital proctoring. A major gap identified was the lack of lightweight, resource-efficient models for low-end devices, which creates access inequality. The paper stressed the need for fairness-aware AI to reduce bias [3].

Lim J (2023) – "Student Acceptance and Compliance in AI-based Online Proctoring" This

study focused on student acceptance, finding that concerns about privacy, fairness, and autonomy outweighed the security benefits of proctoring. It argued that user trust is essential for any system's success and recommended opt-in architectures, though its findings were limited by reliance on self-reported data [4].

Santos M (2023) – "Proctor SU: A Multi-Modal AI Framework for Online Exam Integrity" This paper introduced "Proctor SU," a multi-modal AI framework combining face detection, gaze tracking, and YOLOv3-based object detection. A pilot deployment demonstrated strong performance on standard laptops, validating the feasibility of integrated AI pipelines for strengthening exam integrity, although it was limited to a single implementation [5].

Khan I (2023) – "YOLO-based Real-Time Cheating Detection in Online Proctoring" This work proposed a YOLOv5-based model for the real-time detection of visible cheating cues like mobile phones or multiple people. While achieving strong precision and recall, its primary limitation was its inability to detect non-visual misconduct such as screen switching, whispering, or online collaboration [6].

Lee C (2024) – "Advancing Academic Misconduct Detection with YOLOv8 and Custom Datasets" This research advanced proctoring by developing a YOLOv8 framework trained on a custom dataset of nuanced cheating behaviors like whispering and suspicious glances. This action-recognition approach showed superior performance, but the manual effort required for dataset creation posed significant scalability challenges [7].

Patel R (2023) – "Keystroke Dynamics for Continuous Authentication in Online Exams" This study explored keystroke dynamics as a privacy-friendly method for continuous authentication. By analyzing typing rhythms with machine learning, the system could verify a user's identity throughout an exam without relying on cameras or microphones. Its main limitation was its applicability, being effective only for text-heavy exams [8].

Zhang Y (2024) – "Building a Standardized Fixed-Text Keystroke Dataset for Authentication Research" This paper addressed a key research gap by developing and releasing a standardized, public fixed-text keystroke dataset. This allows for fairer, more consistent comparisons between authentication models, though the fixed-text format may not fully capture the variability of real-world, free-text typing [9].

Chen L (2024) – "Red Teaming Online Proctoring Systems: Identifying Vulnerabilities and Attack Patterns" This study used a red-teaming approach to identify and categorize system vulnerabilities, such as the use of virtual webcams and screen mirroring. It concluded that single-modality systems are highly vulnerable and recommended multi-modal fusion for defense, though the study was diagnostic and did not implement these defenses [10].

Existing Challenges and Gaps: Despite these advancements, current online proctoring systems face several persistent challenges that limit their overall effectiveness, accessibility, and user experience:

- **False Positives:** Many heuristic-based systems or those sensitive to environmental factors (like lighting, background changes) can frequently flag legitimate student behaviors as suspicious, leading to undue stress and administrative burden.

- **Computational Overhead:** Comprehensive multi-modal systems often demand substantial processing power, limiting their deployment in resource-constrained environments or their ability to scale efficiently for large cohorts of students.

- **Real-time Intervention:** While many systems detect suspicious activity, few offer immediate, non-intrusive real-time feedback or automated preventative measures that can deter cheating in the moment. The reliance on post-analysis means misconduct is often addressed after the fact.

- **Limited Integration:** Many solutions operate as standalone systems, lacking seamless integration with common Learning Management Systems (LMS) or enterprise platforms, which complicates deployment and data management for institutions.

- **Privacy Concerns:** While not explicitly detailed in all reviewed papers, the extensive monitoring capabilities of these systems inherently raise privacy concerns that need careful consideration in design and implementation.

How proctonix AI Addresses These Issues: Proctonix AI seeks to overcome these limitations by combining a streamlined, efficient approach with a robust set of real-time detection features, leveraging a demo-focused tech stack to prove concept validity:

- By employing a combination of OpenCV and MediaPipe FaceMesh, Proctonix AI aims for accurate and real-time facial and gaze analysis, reducing false positives through precise landmark detection and custom heuristic logic, without requiring complex calibrations common in older systems.

- The system prioritizes a browser-based frontend and a lightweight Flask backend, ensuring accessibility and reducing hardware demands on student devices, making it less resource-intensive than many multi-object detection pipelines.

- The integration of Web Speech API for voice alerts provides immediate, non-intrusive feedback to students, acting as a real-time deterrent that is less disruptive than manual interventions or post-exam reviews.

- Proctonix AI focuses on a live admin dashboard that enables proctors to monitor multiple students simultaneously, offering real-time alerts and auto-screenshot logging for actionable evidence, enhancing the proctor's ability to intervene effectively.

- While initially focused on core functionalities for a demo, the clear modular design with defined AI processing components lays the groundwork for future scalability and integration with broader LMS platforms, addressing the integration gap observed in current solutions.

III. METHODOLOGY

This section explains how we designed and built the Proctonix AI system. We followed a step-by-step plan to make a strong online exam monitoring tool. Our main goal is to use computer vision along with a fast, web-based system to detect cheating behaviors accurately, while using as little computer power as possible.

A. System Architecture

The Proctonix system is engineered on a three-tier, client-server architecture, a design paradigm chosen for its modularity, scalability, and maintainability. This structure's first tier is the **student-side frontend**, a browser-based application built with HTML, CSS, and JavaScript. Its primary responsibility is to capture and securely transmit a real-time stream of the student's screen activity, microphone audio, and webcam footage to the server. This component serves as the raw data input for the AI analysis pipeline and is the main point of interaction for the test-taker, beginning with a secure login interface.

The intellectual core of the system resides in the **server-side backend**, which is developed using Python and the Flask micro-framework. This tier receives the data streams from the client, executes the AI models for behavioral analysis, and maintains a secure log of all detected incidents. The final tier is the **Proctor Interface**, an administrative console that provides exam supervisors with real-time alerts when suspicious activity is detected. It displays comprehensive incident logs and generates violation reports, enabling proactive intervention through an efficient, alert-driven summary of proctoring events rather than requiring constant monitoring of raw video feeds.

B. Technology Stack

The project's technology stack strategically combines standard web technologies with specialized computer vision libraries to balance accessibility with advanced functionality. The frontend is built with HTML, CSS, and JavaScript, utilizing modern browser APIs for real-time data capture. Specifically, **WebRTC** enables efficient, low-latency webcam and audio streaming, while the **Screen Capture API** monitors screen activity without requiring external software installations. The backend is powered by **Python** and the **Flask** framework, providing a scalable and secure foundation for server-side logic and data handling.

The AI processing pipeline, operating on the backend, relies on **OpenCV** for fundamental image and video processing tasks. This is augmented by **MediaPipe FaceMesh** a high-fidelity, pre-trained model that accurately tracks facial landmarks in real-time, which is critical for gaze and presence detection.

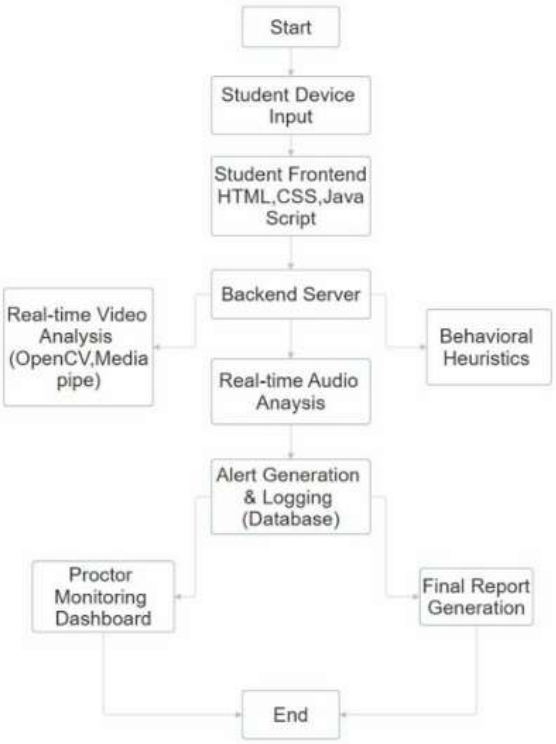

Fig. 1. Workflow of Protonix AI

The system's intelligence is driven by a custom rule-based engine that translates the precise data from these libraries—such as head pose vectors and eye coordinates—into actionable insights. This logic effectively identifies suspicious patterns, such as prolonged off-screen gazes or the detection of multiple faces in the webcam's view.

C. Implementation and Workflow

The operational workflow of the Proctonix system begins when a student authenticates through a dedicated login page and grants browser permissions for their webcam, microphone, and screen. Following a successful identity verification and liveness check, the system establishes a secure, real-time data stream to the backend server. There, the core AI processing pipeline concurrently analyzes the incoming data: **OpenCV** and **MediaPipe** are used to extract detailed facial landmarks for tracking head pose and eye movements, while the audio stream is monitored for unusual sounds. A custom rule-based engine then evaluates this information, flagging behavioral deviations such as

prolonged off-screen gazes or the presence of an additional person.

Upon the detection of a suspicious event, the system immediately generates a real-time alert for the proctor, captures a timestamped screenshot of the incident, and securely logs the event in a database for review. Proctors can monitor these logged alerts through an administrative console. At the conclusion of the exam, the system compiles all flagged incidents into a comprehensive violation report. This final report provides a detailed summary for administrative review, facilitating informed and evidence-based decision-making regarding potential academic misconduct.

D. Reliance on Pre-trained Models and Heuristic Logic

The approach used in this project is based on using pre-trained, cutting-edge computer vision models rather than building a new custom dataset and training new models. We make use of the stability and excellent performance of pre-existing libraries, particularly MediaPipe FaceMesh, which has been pre-trained to recognize facial landmarks in real time. With this strategy, we can concentrate our development efforts on incorporating these potent tools with our unique logic. Our in-house heuristic engine, which converts the unprocessed data from MediaPipe and OpenCV into useful insights about student behavior, is the brains behind our system. As stated in the evaluation section, the system's efficacy is thus assessed based on its capacity to consistently identify and report questionable activity in a real-time, simulated testing environment rather than a fresh dataset.

E. Future Directions and Enhancements

While the current version of the Proctonix system provides a robust framework for automated proctoring, our future work will concentrate on several key areas to enhance its capabilities and address existing limitations. A primary focus will be on transitioning from a rule-based heuristic engine to a more sophisticated, machine-learning-based behavior analysis module. This would involve collecting a new, anonymized dataset of both compliant and non-compliant student behaviors to train a custom deep learning model. Such a model could learn to identify more complex and subtle forms of cheating that a rule-based system might miss, such as a student's nervous tics or a specific pattern of mouse movements.

Furthermore, we plan to explore the integration of multimodal data analysis, combining video, audio, and keystroke dynamics to create a more holistic profile of student behavior. By analyzing keystroke patterns for irregularities and cross-referencing this data with visual and audio cues, we can develop a more resilient and difficult-to-circumvent proctoring solution. These enhancements are aimed at improving the system's accuracy, reducing false positives, and ultimately creating a more equitable and trustworthy online examination environment.

F. Ethical Considerations and Limitations

Proctonix AI is being developed with a strong commitment to student privacy and ethical standards, ensuring all test data is anonymized where possible and stored securely with restricted access. We acknowledge the system's limitations, such as the potential for false positives from normal student behaviors, and position it as a supportive tool for human proctors rather than a final decision-maker. To foster trust and fairness, the system prioritizes transparency by clearly informing students about what data is collected and how it is used. Future work will focus on reducing false positives and refining behavioral analysis to lessen student stress. Ultimately, Proctonix AI aims to balance academic integrity with student welfare, setting a benchmark for ethical, equitable, and effective online proctoring.

IV. RESULTS AND DISCUSSION

The study's primary objective was to evaluate the Proctonix AI system's real-time functional effectiveness. Since the system operates without pre-existing datasets, the evaluation focused on demonstrating its capabilities through key visual outputs and a qualitative assessment. The system's effectiveness is shown by its ability to combine a browser-based frontend that captures video, audio, and screen activity with a backend utilizing **OpenCV** and **MediaPipe FaceMesh**. This allows for the real-time detection of behavioral cues associated with cheating, such as prolonged eye aversion or the presence of additional people. This real-time processing enables immediate alerts, allowing proctors to intervene proactively, which is a significant advantage over post-examination review methods.

Fig 2 :Image of Login Page

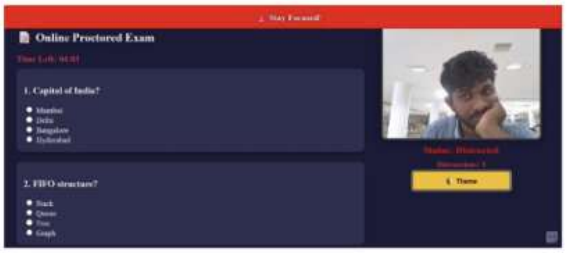

Fig: Output as Distracted

A key finding was the system's resilience and practical efficiency. Proctonix maintains functionality across varying

video feeds and lighting conditions, ensuring an equitable experience for all students, a direct result of using robust, pre-trained models like MediaPipe. Beyond its adaptability, the system's streamlined, browser-based architecture minimizes computational overhead, making it a scalable and viable solution for educational institutions with different levels of technological infrastructure. This focus on accessibility ensures that the system can be implemented for a large number of students without requiring expensive hardware or excessive network bandwidth, extending its benefits to a wider academic community.

Fig: Output as Focused

Finally, a qualitative assessment highlighted the system's user-friendly and ethical design. Proctonix was designed to be non-intrusive, using features like the Web Speech API for unobtrusive voice notifications that gently redirect a student's focus. This approach, combined with a clear data management policy, positions the system as a responsible alternative to more invasive monitoring technologies. Crucially, it is intended to be a tool for human oversight, not a replacement, which addresses the inherent limitations of automated detection and reduces the potential for false positives. This framework ensures a balance between upholding academic integrity and protecting student privacy and well-being.

V. CONCLUSION AND FUTURE WORK

This study successfully developed and evaluated the Proctonix AI system, demonstrating its effectiveness as a real-time, non-intrusive online proctoring solution. The evaluation confirmed that the system's architecture, which combines a streamlined browser-based frontend with a backend utilizing **OpenCV** and **MediaPipe**, is both resilient and scalable. It functions reliably across varied conditions without requiring high-end hardware, making it an accessible tool for a wide range of educational institutions. By providing immediate alerts for proactive intervention, Proctonix serves as a powerful and efficient **tool for human proctors**, effectively balancing the critical need for academic integrity with a strong commitment to student privacy and well-being.

Future development will focus on enhancing the sophistication and fairness of the AI detection engine. A primary objective is to **reduce false positives** by developing more complex behavioral analysis models that can better understand the context of a student's actions, distinguishing between innocuous movements and genuine misconduct. Research will also explore the integration of more advanced machine learning models to detect a wider range of nuanced behaviors. Furthermore, we plan to expand on the system's ethical framework by incorporating enhanced transparency features, such as a student-facing dashboard that provides clear information on the data being monitored, thereby strengthening user trust and setting a higher standard for responsible proctoring.

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ACCEPTANCE LETTER

Manuscript ID : **ICACRS291**
Manuscript Title : **Proctonix AI - Smart Exam Proctoring Using AI and Webcam**
Author/s : **Vishnu Priya,A Govardhan Naidu,I Mukesh Manikanta,Ch Santosh Kumar,B Issac Preetham**

Dear Author/s,

Greetings from Mount Zion College of Engineering and Technology (MZCET)!

We are pleased to inform you that your research manuscript, as mentioned above, has been accepted for presentation at the International Conference on Automation, Computing and Renewable Systems (ICACRS 2025). The conference will be held from December 10–12, 2025, at Mount Zion College of Engineering and Technology, Pudukkottai, Tamil Nadu, India.

You have been selected to deliver an oral presentation of your research work at the ICACRS 2025 conference. This international event brings together leading researchers, academicians, and industry professionals to share their insights and foster innovation in the fields of automation, computing, and renewable systems.

In this regard, we kindly request you to submit the final manuscript, copyright form, and other required documents to the conference email at the earliest to facilitate timely publication of your paper. When submitting your final manuscript, please ensure that all revisions are clearly highlighted as per the reviewer comments provided.

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